

Program overview

20-May-2025 12:52

Year 2024/2025
 Organization Mechanical Engineering
 Education Master Materials Science and Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master MSE 2024							
Master Year 1							
MSE Core Programme (obligatory for all students: 41 ECTS)							
CH3531	Functional Ceramics	3					
MS43006	Structure and Properties of Materials	6					
MS43011	Characterisation of Materials	5					
MS43016	Metals Science	3					
MS43020	Computational Materials Science	3					
MS43025	Mechanical Behaviour of Materials	4					
MS43030	Processing of Materials	3					
MS43035	Thermodynamics and Kinetics of Materials	3					
MS43045	Societys Needs: Case Studies and Materials Challenges	4					
MS43050	Polymer Science	4					
WM0320TU	Ethics and Engineering	3					
MSE Specialisation Courses							
MSE Specialisation Materials in Engineering Applications							
MSE MEA Obligatory fixed (6 ECTS)							
MS43100	Science of Failure	3					
MS43120	Corrosion Engineering	3					
MSE MEA Obligatory selection (Select at least 8 ECTS)							
MS43110	Joining Technologies	3					
MS43115	Materials Selection for Engineering Applications	3					
MS43210	Advanced Characterisation	4					
MS43315	Recycling Engineering Materials	4					
MSE Specialisation Metals Science and Technology							
MSE MST Obligatory fixed (7 ECTS)							
MS43100	Science of Failure	3					
MS43200	Metals Science II	4					
MSE MST Obligatory selection (Select at least 7 ECTS)							
MS43205	Computational Materials Science II	3					
MS43210	Advanced Characterisation	4					
MS43215	Steel Science	3					
MS43220	Corrosion Science	3					
MS43330	Sustainable Metals Production	4					
MSE Specialisation Materials for Sustainable Development							
MSE MSD Obligatory fixed (8 ECTS)							
MS43305	Materials for Hydrogen and Solar Energy Technology	4					
MS43315	Recycling Engineering Materials	4					
MSE MSD Obligatory selection (Select at least 6 ECTS)							
MS43205	Computational Materials Science II	3					
MS43310	Materials at High Temperature	4					
MS43330	Sustainable Metals Production	4					
MSE Free Specialisation 14 EC (to be filled in consultation with master coordinator)							
MSE Recommended electives							
AE4ASM001	Design of lightweight structures I: Composites & Metals	3					
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3					
AE4ASM005	Fatigue of Structures & Materials	3					
AE4ASM103	Functional Coatings	3					
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5					
AE4ASM504	Structural Integrity and Maintenance	3					
AE4ASM510	Design & Analysis of Composite Structures II	3					
BM41035	Biomaterials	4					
CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
MS43040	Machine Learning for Materials Design	4					
MS43100	Science of Failure	3					
MS43110	Joining Technologies	3					
MS43115	Materials Selection for Engineering Applications	3					
MS43120	Corrosion Engineering	3					
MS43200	Metals Science II	4					

MS43205	Computational Materials Science II	3	
MS43210	Advanced Characterisation	4	
MS43215	Steel Science	3	
MS43220	Corrosion Science	3	
MS43305	Materials for Hydrogen and Solar Energy Technology	4	
MS43310	Materials at High Temperature	4	
MS43315	Recycling Engineering Materials	4	
MS43325	Application of Materials in High Tech Engineering	3	
MS43330	Sustainable Metals Production	4	
MS43805	Metallic Materials in Maritime Structures	5	
MS43810	Materials in Art and Design	3	
MS43825	Radiation Damage in Materials	4	
SET3055	Economics and Regulations of Sustainable Energy Systems	4	
Master Year 2			
MSE MSc Projects and Assignments			
IFM4040	Joint Interdisciplinary Project	15	
MS43702	Materials Elective Project I	2	
MS43706	Materials Elective Project II	6	
MS53200	External Research Project	15	
MSE MSc Graduation Project (45 ECTS)			
MS53010-20	MSE Literature Research	10	
MS53035	MSE MSc Thesis	35	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

Master MSE 2024	
Director of Studies	Y. Gonzalez Garcia
Program Title	Master Materials Science and Engineering
Program Coordinator	The coordinator is the person for questions or problems related to the individual study programme and for monitoring progress. The coordinator can be contacted through the email: MSEcoordinator-ME@tudelft.nl
Prerequisites	Dr. Yaiza Gonzalez-Garcia (Program Coordinator and Director of Studies) Saskia van der Meer (Education Assistant) There are several ways to be admitted to the MSc program Materials Science and Engineering: - As a continuation of an academic BSc program, either in the Netherlands or abroad. - After completing a bachelor's program at a Dutch polytechnic institute or the Netherlands Defence Academy (NLDA). More details about the admission and procedures are found in: https://www.tudelft.nl/onderwijs/opleidingen/masters/mse/msc-materials-science-engineering/admission-and-application
Introduction 1	Materials Science & Engineering (MSE) is an interdisciplinary field focusing on the relationship between structures and the behaviour of materials (science) and the sustainable design of these structures to produce predefined properties for specific applications (engineering). Students will consider materials over a wide range of length and time scales from the behaviour of collections of individual atoms to macroscopic structural and functional properties and from raw material processing through sustainable production and application of materials, to failure, durability and recycling. Students will learn to understand the behaviour of materials under different conditions and assess their suitability in products and industrial processes. Moreover, students will study how to design material properties at nano and micro levels in a manner that is suitable for applications on different scales. The two-year program (120 EC) is taught in English and incorporates academic learning and active research. Course modules are delivered through lectures, lab sessions, workshops, and seminars.
Program Goals	The aim of the program is to offer students high-quality, interdisciplinary education, and to turn out graduates capable of making immediate and significant contributions to a wide range of industrial and academic areas at a global level. The main goals are: 1. to provide students with a sound and thorough understanding of the underlying scientific and engineering principles involved in Materials Science and Engineering 2. to enhance knowledge of sustainable materials design, selection, processing and characterisation 3. to build up awareness of the environmental, economic and human aspects of materials selection, usage, recycling and disposal.
Exit Qualifications	A graduate in Materials Science & Engineering meets the following qualifications: 1. Profound knowledge of Materials Science and Engineering and the underlying engineering sciences (mathematics, physics and chemistry) 2. The capability to apply this advanced knowledge effectively to reach the forefront of scientific or industrial research and development. 3. The skills to actively and independently apply this knowledge for analysing, modelling, simulating, designing and performing research with respect to problems related to Materials Science and Engineering 4. The capability to solve technological problems in a systematic way. Professional attitude towards identifying and acquiring lacking expertise, and monitoring and critically evaluating existing knowledge. 5. The capability to work both independently and in multidisciplinary teams. Effective communication with both professionals and a non-specialised public. 6. The capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility regarding sustainability, economy and social welfare.
Program Structure 1	The program is structured as followed: - Core Program (41 EC) obligatory set of courses which provides the basic knowledge required by all graduates in Materials Science and - Specialisation (34 EC) focusing on a selected area of interest - Literature Research (10 EC) - Master Thesis Project (35 EC): independent scientific investigation that leads to a thesis. The specialisation focuses on science, engineering and application-related issues based on the interest of the student. The pre-defined specialisations are: 1. Materials in Engineering Applications (MEA) 2. Metals Science and Technology (MST) 3. Materials for Sustainable Development (MSD) 4. Self-defined (SDS) The student can also define their own specialisation upon approval by the program coordinator and the Board of Examiners.

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

Master Year 1

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE Core Programme (obligatory for all students: 41 ECTS)	
Introduction 1	Core Programme The Core Programme, that runs through the first semester and a small part of the second semester, contains identical modules for all students. It focuses on Materials Science and Engineering fundamentals and general knowledge. It offers students a firm grounding upon which they can build more specialised or wider ranging interests. This core programme comprises the compulsory modules shown below (total of 41 EC).

CH3531	Functional Ceramics	3
Responsible Instructor	Dr. E.M. Kelder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc: general and inorganic chemistry, basic quantum chemistry	
Course Contents	Ceramic materials show a wide variety in chemical, optical and electrical properties. They are used in many different applications, ranging from catalysts, fuel cells, batteries and sensors to solar cells, photo-catalysts and luminescent nanoparticles. This course will treat the basic properties of functional ceramic materials. The following topics will be included: - crystal structures - bonding in inorganic solids - basic electronic structure and properties of semiconductors - defect chemistry including point defects, non-stoichiometry and doping - ionic and electronic charge transport - semiconductor junctions and space charges.	
Study Goals	After this course, the student is able to: - Correlate crystal structure and band structure to optical and electrical properties of ceramic materials - Describe the operating principles of simple devices based on semiconductor junctions (solar cells, diodes, photo-electrodes) and ionic conduction (Nernst sensors, (Li-ion)-batteries, solid oxide fuel cells (SOFCs), and catalysts) - Identify and predict relationships between point defects, dopants, space charges, and ionic/electronic conduction in both semiconductors and ionic conductors, and perform simple calculations	
Education Method	Lectures, self study, instruction classes	
Books	Optional: - "Physical Ceramics" by Y-M Chiang, D. P. Birnie, and W.D. Kingery, John Wiley & Sons, ISBN: 0-471-59873-9	
Reader	- Several handouts and the lecture slides will be made available on Brightspace	
Assessment	The exam will be a closed book exam.	
Permitted Materials during Tests	- Calculator, overview of most important equations will be handed out during exam, the reference sheets.	

MS43006	Structure and Properties of Materials	6
Responsible Instructor	Dr.ir. M.H.F. Sluiter	
Instructor	Dr. M.W.E.M. Alfeld	
Instructor	Dr. P. Dey	
Co-responsible for assignments	Dr. P. Dey	
Co-responsible for assignments	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	2.4.0.0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Required for	most other courses in the materials science curriculum	
Expected prior knowledge	THE LIST BELOW IS VERY IMPORTANT! IT IS THE STUDENTS' RESPONSIBILITY TO START WITH PROPER PRIOR KNOWLEDGE. vectors & matrices (basics of linear algebra) complex numbers (square root of -1 etc. etc.) calculus Fourier transform two pdf texts with details and a self-test are available under "content" on Brightspace: MatSci - preknowledge mathematics MatSci - entry level test mathematics high school level chemistry, e.g. (but not limited to) -basic understanding of the periodic table of the elements -concept of valence -chemical bond high school level physics, e.g. (but not limited to) -concepts of energy, potential energy, kinetic energy -Newton's laws -electrostatics -law of gravitation -laws of thermodynamics	
Course Contents	part 1 (1st quarter taught by Marcel Sluiter) Crystals and Crystallography (chapter 4 Naumann textbook) Crystals (primitive translations, unit cell, lattice points, basis, Miller (Bravais) indices, interplanar spacing) Crystal Systems and Symmetry Structural Relationships (density and packing, atomic density, mass density, ...) Interstices Quasicrystals The Structure of Matter (chapter 5 Naumann textbook) metals Intermetallic Compounds Ionic Compounds Covalent Structures Structure of Glass Structure of Polymers Quantum Mechanics (chapter 2 Naumann textbook) Atomic Structure The Electron (de Broglie, Heisenberg uncertainty) Schrödinger Wave Equation One Electron Approximation Periodic Table Chemical Bonding (chapter 3 Naumann textbook) Ionic Bonding Covalent Bond Metallic Bond Atomic and Ionic Radii Secondary Bonding Other Potential Functions ----- mid term exam over material covered in part 1 ----- Part 2 (2nd quarter taught by Poulumi Dey and Matthias Alfeld) Physics and Chemistry of Materials Ch15 -Ch22 Naumann covering: Distribution Functions Lattice Vibrations and Phonons Thermal Properties of Solids Free Electrons in Metals Band Theory of Metals Semiconductors Theory and Application of Junctions Transistors, Quantum Wells and Superlattices	
Study Goals	The student is able to rationalize materials properties and relate them to atomic and electronic structures.	

Education Method	Q1 part: lectures and practice sessions Literature and Study Materials Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X (Chapters 1-5) additionally some other course material will be offered in the form of pdf files during the course. Q2 part: lectures and practice sessions. Literature and Study Materials Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X (Chapters 15-22)
Computer Use	microsoft office (word, powerpoint, excel) or equivalent simple text editor
Literature and Study Materials	Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X additionally some other course material will be offered in the form of pdf files during the course.
Books	Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X
Assessment	Written exams, on-campus (possibly shifted to online, depending on the Covid safety situation in Q1 part 1 and Q2 part 2). Concerning Part 1: Closed-book multiple choice questions and numerical questions. Concerning Part 2: On-campus exams will be closed book with a focus on multiple choice questions. Online exams will be open book with open questions. Resits in Q2 (part 1) and Q3 (part 2) under identical conditions.
Permitted Materials during Tests	generally closed-book exams
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Tags	Algebra Analysis Calculus Chemistry Energy Physics Quantum
Contact	part 1: Marcel Sluiter: M.H.F.Sluiter@tudelft.nl part 2: Matthias Alfeld: M.Alfeld@tudelft.nl

MS43011	Characterisation of Materials	5
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the Characterisation of Materials course, general research tools in materials science are introduced. The following modules constitute the course, although the number and content of modules may be slightly modified upon starting of the course. If that is the case, students will be informed through Brightspace. - Module 1: Optical Microscopy - Module 2: Electron Microscopy - Module 3: Thermal analysis - Module 4: Mechanical Behaviour - Module 5: Electrochemistry - Module 6: X-ray diffraction	
Study Goals	On completion of the course, the student will be able to select techniques for materials characterisation, describe those techniques, specify the physical principles involved and interpret the results obtained. More specifically, the students will be able to: - use optical microscopy for materials characterisation. This includes sample preparation and selection of appropriate etching agents, - describe the principles and justify the use of scanning electron microscopy and energy dispersive x-ray spectroscopy in materials research and analyse the data, - derive information on phase transformations from dilatometry experiments, compare the results to CCT diagrams and correlate the data with the observed microstructure and the phenomena occurring during welding and heat treatments, - select appropriate mechanical tests and analyse the data, - identify corrosion mechanisms and quantify parameters which influence the corrosion rate, - describe the principles of the x-ray diffraction technique and analyse the data, - communicate in writing the experimental condition of the specific techniques and results obtained. Write different type of reports as requested, per lab class. Small modifications in the content and number of modules may lead to a slight modification and/or reduction of the goals above. In that case, students will be informed via Brightspace.	
Education Method	Lectures and lab sessions.	
Literature and Study Materials	Literature, readers and study materials will be provided in the course.	
Assessment	The assessment of the course includes: - Group assignments (70% of the grade). - An individual open-book multiple-choice exam (30% of the grade). Students are required to obtain a minimum mark of 6 for both parts in order to pass the course. Peer-reviewing will be considered in the final grade (+/- 1 point) The online course includes individual questionnaires and quizzes for self-assessment that will not contribute to the grade.	
Permitted Materials during Tests	During the open-book multiple choice examination, it is permitted to use any material from the course. The student can use laptop or tablet to access course material. However, the access to internet and the use of telephone is not allowed. It is strongly advised that the student brings electronic devices fully charged to the exam, since in some classrooms the access to electric sockets may be limited.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43016	Metals Science	3
Responsible Instructor	Prof.dr.ir. L.A.I. Kestens	
Instructor	Ir. N. Geerlofs	
Contact Hours / Week x/x/x/x	0.4.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Metals represent a vital class of materials for a technological society. This course examines the structure and properties of metals across a range of length scales, addressing issues of microstructural changes and phase transformations, metals production techniques and the behaviour of metals in generic applications. The course covers microstructures, mechanical properties in relation to microstructures and crystallographic aspects of physical metallurgy. Microstructural aspects include: 1) basic elements of the plasticity of metal alloys including concepts such as the theoretical strength of single crystals, plasticity by dislocation motion in single crystals, stress fields induced by dislocations, dislocation energy, force on dislocations (Peach-Koehler law), multiplication of dislocation (Frank-Read source). 2) Strengthening mechanisms: work hardening (Taylor equation), grain-boundary strengthening (Hall-Petch relation), solid solution strengthening, precipitation hardening, age-hardening. 3) Various stages of work hardening. Dynamic recovery, effect of stacking-fault energy. Shockley partial dislocations. 4) Basics of crystal plasticity theory: Taylor vs Sachs theory (introduction). 5) The crystallographic theory of twinning. 6) the phenomenological theory of martensitic transformations, occurring under either thermal or mechanical driving force. 7) the origin of crystallographic texture in metallic microstructures, the representation of texture and the experimental techniques to measure texture on a macro- or microscopic scale. The typical textures of steel and aluminium alloys (rolling, annealing and transformation textures). 8) The deformation microstructure: cell formation, deformation bands, shear bands, stored energy of plastic deformation 9) Thermally activated recovery mechanisms: recovery (formation of low-angle grain boundaries), recrystallization (nucleation and growth), JMAK equation, curvature driven grain growth (Gibbs-Thompson relation), Zener pinning.	
Study Goals	1. Understand the basic deformation mechanisms of materials. 2. Acquiring knowledge about the mechanical behaviour of materials in relation to the fundamental deformation mechanisms. 3. To designate elements in the material design that affect strengthening behaviour. 4. Acquiring knowledge concerning the crystallographic structure and properties of mechanical twins. 5. To gain insight in the process of martensitic phase transformation. 6. Being able to establish a link between the typical characteristics of the (sub-)structure of a plastically deformed metal and the properties of the recrystallised structure after annealing. 7. Mastering the basic concepts of quantitative texture analysis and being able to identify and analyse the crystallographic texture of standard steel and aluminium grades.	
Education Method	Standard white board lectures (either on-line or on-campus, pending on faculty regulation) backed-up by slides or textbooks. Q&A sessions to apply the lectures in problem solving exercises.	
Assessment	Standard: on-campus exam (closed book) with open questions or application oriented questions; If exam is online (because of faculty regulation): open book examination with application oriented questions only. Typical duration of exam: 3h-3.5h	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	

MS43020	Computational Materials Science	3
Responsible Instructor	Dr.ir. M.H.F. Sluiter	
Co-responsible for assignments	Dr.ir. C. Bos	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MS43205 Computational Materials Science 2.	
Expected prior knowledge	Basic knowledge of algorithms, basics of computer programming, a working knowledge of PYTHON, basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schroedinger eqn, uncertainty principle); basics of thermodynamics; calculus. Also some familiarity with basic materials science concepts such as crystal structures - translations & periodicity, crystal basis, vacancy, dislocation, grain boundary, grain coarsening, phase transformation such as solidification, spinodal decomposition, nucleation. If you have no prior experience with Python you can follow a self-study course at https://www.learnpython.org/ (of course, you may also use other Python courses/resources to prepare yourself.) Take note: * MS43020 requires that you can use Python to write simple computer programs BEFORE you start the course. * Before you start the course you must verify your Python skills by taking a test (location to be specified) Make this test in Q1, so that you may remedy your lack of skills BEFORE Q3. NB It is NOT recommended to skip this course in the 1st year of the MSc because then you will have two chances at passing the exam only. Much better is to prepare in a timely fashion and take part in the course from the start.	
Course Contents	Parts 1 and 2 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8). Home work and many of the illustrations utilize Jupyter notebooks. Modeling and simulation What is meant by computational materials science and engineering? Scales in materials structure and behavior How to develop models The random-walk model Random-walk model of diffusion Connection to the diffusion coefficient Bulk diffusion A random-walk simulation Random-walk models for materials Simulation of finite systems Sums of interacting pairs of objects Perfect crystals Cutoffs Periodic boundary conditions Implementation Long-ranged potentials Electronic structure methods Quantum mechanics of multielectron systems Early density functional theories The Hohenberg-Kohn theorem Kohn-Sham method The exchange-correlation functional Wave functions Pseudopotentials Use of density functional theory Interatomic potentials The cohesive energy Interatomic potentials Pair potentials Ionic materials Metals Covalent solids Systems with mixed bonding What we can simulate Determining parameters in potentials Molecular dynamics Basics of molecular dynamics for atomic systems An example calculation Velocity rescaling Molecular dynamics in other ensembles Accelerated dynamics Limitations of molecular dynamics Molecular dynamics in materials research The Monte Carlo method Ensemble averages The Metropolis algorithm The Ising model Monte Carlo for atomic systems Other ensembles Time in a Monte Carlo simulation Assessment of the Monte Carlo method	

	Uses of the Monte Carlo method in materials research
Course Contents Continuation	computational materials science II (MS43205) is a follow-up course. In the follow-up the 2nd part of the textbook by LeSar is used.
Study Goals	After passing the course, the student will be able to... 0) can write a piece of PYTHON code that represents a model of materials behavior 1) determine what type of computational model is needed to simulate a particular phenomenon in materials science 2) understand the basic principles behind several classes of models, such as stochastic models involving random-walk, and Monte Carlo sampling simulations involving finite systems, e.g. supercell modelling as used in many ab initio electronic structure methods and molecular dynamics 3) understand, apply, interpret and analyze results from a selection of models: - diffusion through random walk - interatomic interactions in a lattice gas through supercell calculations using a electronic density functional method also called "ab initio method" - Monte Carlo sampling of the partition function - force fields as a glue between ab initio and empirical information on the one side and dynamics over extended time and length scales on the other side - Molecular dynamics simulations involving force fields 4) visualize data for insight and for communication
Education Method	Lectures & homework computer exercises (TU Delft computers can be used)
Computer Use	Yes, for exercises you can use TU Delft computers.
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8).
Prerequisites	* MS43020 requires that you can use Python to write simple computer programs BEFORE you start the course. * Before you start the course you must verify your Python skills by taking a test (location to be specified) Make this test in Q1, so that you may remedy your lack of skills BEFORE Q3. If you have no prior experience with Python you can follow a self-study course at https://www.learnpython.org/ (of course, you may also use other Python courses/resources to prepare yourself.)
Assessment	Computer-based exam. The exam consists of exercises where you write Python code to simulate phenomena in materials - similar in style and format to the homework exercises during the course.
Permitted Materials during Tests	TUD PC.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Tags	Algorithms Analysis Artificial intelligence Calculus Computer Engineering Modelling Numeric Methods Programming Programming concepts Video
Department	ME Department Materials Science & Engineering

MS43025	Mechanical Behaviour of Materials	4
Responsible Instructor	Dr. S. Kumar	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the essential knowledge required to understand and predict the mechanical behaviour of materials. The following aspects are addressed and illustrated with reference to a specific material class (in brackets): - Tensors and kinematics - Generalized elasticity - Viscoelasticity (polymers) - Plasticity (metals) - Finite-strain hyperelasticity (rubber/biological tissues)	
Study Goals	Upon completion of the course, students should be able to: 1. Model elastic behavior of isotropic and anisotropic materials 2. Model time dependent elastic behavior of polymers 3. Model metal plasticity using yield surfaces such as Tresca and Von Mises 4. Model behavior of rubber and biological tissues under large strains	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture notes will be provided on Brightspace. Additional reading material: "The Mechanics of Constitutive Modeling" by Niels Ottosen, Matti Ristinmaa Note that above reading is optional/supplemental and not a textbook for this course.	
Assessment	Two homework assignments will be given at the mid-point and end of the course. Each assignment will account for 10% of the final grade (cumulatively, 20%). The rest 80% of the final grade will be assessed by a written final examination. Note: Exam will be conducted with open lecture-notes. Books and electronic devices are not allowed. Students are advised (not mandatory) to make their own concise formula sheets for the exam. Simple calculators are allowed. Graphing calculators are not allowed.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Mechanics Modelling	
Department	ME Department Materials Science & Engineering	
Contact	Contact by email: Sid.Kumar@tudelft.nl	

MS43030	Processing of Materials	3
Responsible Instructor	Dr. N.D. Dogan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course focuses on quantitative descriptions of momentum and energy transfer applied to different materials processing steps and applications. This course introduces the fundamentals of momentum and energy transfer phenomena for solving process and materials-related problems. The physical mechanism of motion in fluids and heat transfer will be described. The viscosity and thermal conductivity of fluids as a function of temperature will be explained. Besides, the mathematics describing the velocity and temperature distribution of fluids as a function of position and time will be explored. Practical examples of momentum and energy transfers in materials processing will be provided. The following topics will be discussed during the lectures: Momentum transport Viscous properties of fluid Laminar flow and the momentum equation Turbulent flow and complex flows Energy balance applications in fluid flow Energy transport Fouriers law and thermal conductivity Heat transfer and the energy equation Correlations and data for heat transfer coefficient Conduction of heat in solids	
Study Goals	By the end of this course, you should be able to: - Explain basic principles of heat and energy transport - Formulate materials processing problems in terms of mathematical equations - Solve simplified problems in materials processing	
Education Method	Core concepts, theories, and approaches to understand transport phenomena will be covered in lectures. Several examples will be discussed to demonstrate the use of energy and momentum transfer in materials engineering. Teaching material, including a course outline, notes, problems, assignments, and course announcements are available on the Brightspace website.	
Assessment	To successfully complete the course, you will need to score minimum 6 out of 10 points in the final grade. This final grade consists of two homework assignments, a report and presentation. The assignments account for 20% of the total grade. The report contributes to 50% while the presentation contributes to 30% of the total grade.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43035	Thermodynamics and Kinetics of Materials	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course introduces the thermodynamics and kinetics of phase transformations. Thermodynamical subjects include Gibbs free energy, chemical potential, ideal and real solutions, binary phase diagrams, Gibbs phase rule, Gibbs-Thomson effect, ternary systems, and ternary phase diagrams. Kinetics refer to atomic mechanism of diffusion, substitutional and interstitial diffusion, high-diffusivity path, and diffusion in ternary alloys. The different type of interfaces that can be found in a microstructure are also treated in the course (free surfaces, grain boundaries and phase interfaces). The course also introduce mechanisms associated to phase transformations in the solid state, such as nucleation and growth, as well as basic concepts on phase transformations in steels.	
Study Goals	A the end of this course students will be able to: 1. Formulate and apply the thermodynamics of unary systems and of binary systems based on the ideal and regular solution models. 2. Interpret equilibrium phase diagrams and formulate the thermodynamics behind them in terms of Gibbs free energy, enthalpy, entropy and chemical potential. 3. Describe and identify the process of atomic diffusion in crystalline structures. 4. Describe and identify different types of interfaces and estimate their interfacial energies in crystal structures. 5. Relate thermodynamics, diffusional processes, interface types and presence of secondary particles with the occurrence and progress of grain growth and solid-solid phase transformations. 6. Formulate solid-solid phase transformations in terms of nucleation and growth and describe their progress based on TTT diagrams and Avramis model. 7. Describe basic characteristics of solid-solid phase transformations in steels.	
Education Method	Lectures and applied questions.	
Assessment	Close book written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	

MS43045	Societys Needs: Case Studies and Materials Challenges	4
Responsible Instructor	P. Taheri	
Instructor	S.T. Abrahami	
Instructor	Dr. G.H.J. Langejans	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Responsible instructors: Peyman Taheri (coordinator) and Geeske Langejans In this course, you will create a solution or improvement for a material sustainability problem. You will work in groups of four, with groups being created by the lecturers. The course has two parts: 1) the project part, where you will gain insight into the physics and technological aspects of the design and application of a material; and 2) the materials and people part, where you will gain insight into the (historic) societal and socio-economic context of materials and sustainability. As a result, you will gain insight into the physics and technological aspects of the design and application of a material. You will learn about value-sensitive design and the complex interplay between materials and people, including its long, sometimes colonial, history. Each group will choose a material sustainability challenge. In your project, you will research the literature and propose a solution for this challenge. Please note that you are expected to think outside the box and suggest solutions with a reasonable scientific basis. You will conduct a quantitative analysis that takes note of the material's microstructure and the effects of fabrication and processing on the microstructure and, consequently, the performance of the material. You will report on the resources and energy used in producing the properties needed for the required function and long-term performance. Part 1 is mostly self-guided learning with your team. In Part 2, you will conduct a stakeholder analysis relevant to your project and give a personal reflection on the materials, people, and sustainability aspects of your project. Part 2 is mostly group study and done in-class. During the workshops, you will work on assignments, and these assignments will be part of your report. Finally, in two workshops, you will learn and practice giving and receiving feedback; this transferable skill is very useful during (international) teamwork.	
Study Goals	By the end of the course, you have or can: 1. Explain one potential material innovation that could make the problem identified under more sustainable, use the categories processing, structure, properties and performance of materials in your explanation. 2. Reflect on the impact that materials have on people and societys by reviewing a stakeholder analysis and the materials history form. 3. Report and weigh your findings in a scientific oral presentation and written report, and review work by peers.	
Education Method	Lectures/workshops plan You will meet with us/your lecturers almost every week in Block 1 on Monday. The weekly lectures and workshops prepare you for the deliverables. The workshops require some preparation beforehand. Session 1: Introduction to the course and project topics (D1.1 due week 1, D1.2 due week 2) Session 2: workshop working together Session 3: Materials and people, (pre)historic perspective Session 4: Midterm symposium (D2) Session 5: Materials and people, workshop stakeholder analysis Session 6: Materials and people, workshop reflection on historic and colonial perspective Session 7: Final symposium (D4) Session 8: Online submission deadline of report (D5) Progress meetings Your team will also be coached by one of the lecturers. With your coach, you can discuss your process, team dynamics and your project management. The group is responsible to contact the lecturer to set up meeting ahead of time; plan these meetings timely. Reading and sources List of reading materials can be found on Brightspace.	
Assessment	Deliverables These are the items we expect you to deliver during the course: D1.1: Project research question D1.2: Peer review of another groups research question D2: Midterm presentation D3: Final project presentation D4: Project report, including appendices of the stakeholder analysis and materials history form Grading D3: 40% D4: 60% D1, D2 and active participation will greatly contribute to the excellence of D2-D4. In this course you will be assessed along the following 10 criteria: Criterion 1 : Clear goals, context setting and urgency -> research question and motivation. Criterion 2: Content /data/knowledge -> well researched and relevant data. Criterion 3: Well-researched identification of different stakeholders -> relevant people, industries and institutions that are influenced or linked to the suggested material innovation Criterion 4: Reflection and weighing of arguments ->personal reflection in materials and people reflection, reflection on stakeholders, weighing of choices for material innovation, discussion section of report. Criterion 5: Concluding -> logical outcome, answering research question. Criterion 6: Presentation of data -> structure of text and presentation, logical flow from research question to conclusion, use of images and tables, oral presentation skills Criterion 7: Academic skills -> referencing, separating data and opinion. Criterion 8: No plagiarism. Criterion 9: AI reflection. Criterion 10: Sticking to the deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on	

<https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses>.

To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams>.

*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations>.

Department

ME Department Materials Science & Engineering

MS43050	Polymer Science	4
Responsible Instructor	Dr. E. Mendes	
Instructor	Prof.dr. S.J. Picken	
Instructor	Dr. G.A. Filonenko	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, you will be taught the basics of polymer science starting from definitions and various kinds of molecular structures of polymers until their properties in solution, melt and bulk material such as in elastomers, gels and polymer glasses. You will be taught how universal polymer chain theoretical models (ideal chain and excluded volume chain) can describe molecular behavior and chain conformations in bulk or solution for polymers with any chemical structure. Visco-elasticity of polymer solutions in dilute, semi dilute and melt will be considered from non-entangled to entangled regime in solutions and melts. Chain dynamics, reptation theory and tube models will be considered and you will be taught how to interpret experimental data in the light of those theories. You will learn how to extend visco-elasticity predictions to the glassy state and how free volume theories describe a polymer glass. Polymer miscibility and polymer blends, block copolymers self-assembly in solution and in melt and other ordered phenomena in polymers (crystallinity and liquid crystallinity) will also be taught. Finally, main aspects of polymer composites will be taught.	
Study Goals	We will give some examples of how polymers are applied and which one(s) to choose for the purpose. Study Goals Knowledge: 1) be able to explain universal polymer chain models and chain statistics in relation to chemical structure 2) be able to explain what are the main quantities related to polymers such as, persistence length, radius of gyration, reptation time, glass transition temperature, etc. 3) be able to describe how various polymer characterisation methods work Understanding: 4) be able to explain the role of entropy and of excluded volume interactions in polymer conformations and visco-elasticity 5) be able to formulate the concepts of miscibility, solvent quality and phase separation in polymer solutions and polymer blends 6) be able to formulate the concepts of entanglements and polymer tube Analysis (critical thinking): 7) be able to interpret quantitatively (characterization) experimental data on polymer materials 8) be able to calculate basic quantities related to structure and dynamics of polymer materials Application: 9) be able to select or propose a polymer for a given application based on knowledge acquired in the course. 10) have a general overview of what polymers are used for, also discussing some examples.	
Education Method	Lectures	
Literature and Study Materials	Book part 1: Physics of Polymers M. Rubinstein & R. Colby Oxford Book part 2: From Polymers to Plastics A.K. van der Vegt VSSD 2002 Delft University Press	
Assessment	Written exam with open slides.	

WM0320TU	Ethics and Engineering	3
Module Manager	M. Sand	
Instructor	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%, individual grade), presentation and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course WM0329TU (6 ects).	
Elective	Yes	
Category	MSc niveau	

MSE Specialisation Courses

Introduction 1

Specialisation Course

In the second semester, students start to take a Specialisation Course based on their own interests. This course focuses on science, engineering and application-related issues within a specific theme and has a total size of 34 EC. At the end of the first semester students choose their Specialisation Course.

Pre-defined Specialisation Courses

A number of Specialisation Courses have been pre-defined. Each pre-defined Specialisation Course comprises obligatory modules, possibly some modules of which at least a certain number should be chosen, and elective modules (see below).

Elective Modules

An overview of recommended elective modules for each of the pre-defined Specialisation Courses can be found in the respective descriptions. Taking these elective modules will per definition be approved by the Board of Examiners. Other elective modules, however, require explicit approval in order to assure that the acquired knowledge and skills contribute to the exit qualifications of the MSE master's programme. Students should take notice of the following:

- Any of the following items can potentially be regarded as an elective module, provided its level and weight are sufficiently high:
 - a module scheduled at Delft University of Technology;
 - a special assignment or project defined and supervised by a scientific staff member of the Department MSE or someone participating as lecturer in the master's programme MSE;
 - an external research project, organised by an external organisation or institution, supervised by local staff, and approved and assessed by a scientific staff member of the MSE department or someone participating as lecturer in the MSE master's programme.
- Before taking elective modules, each student should present his or her programme to the Board of Examiners for approval. Changes can be allowed at a later stage, but only after renewed approval by the Board of Examiners.
- Cooperation between students during a minor part of an assignment or project is allowed, but each elective module is assessed on an individual basis. This means that each assignment or project should be a unique piece of work and lead to an individual report (and presentation).

Self-defined Specialisation Courses

Students can also choose to define their own Specialisation Course. Examples of themes for self-defined Specialisation Courses are composites, biomaterials, polymers, functional ceramics and materials & nanotechnology. Modules for these self-defined Specialisation Courses can be offered by the MSE department or by other departments at TU Delft.

Students should get approval for self-defined Specialisation Courses by the Board of Examiners. To get this approval students should take notice of the following:

- The theme of the Specialisation Course should not be covered by one of the pre-defined Specialisation Courses;
- At least 14 EC of the modules should form a consistent set related to the theme of the Specialisation Course;
- The majority of the modules should connect to the exit qualifications of the master Materials Science and Engineering;
- Students should write a letter in which they motivate why they want to define their own Specialisation Course and how they will deal with a possible lack of knowledge.

Notes:

- Student activities that are mainly undertaken to acquire work experience or to make a first acquaintance with job conditions cannot be regarded as elective modules.
- The supervisor of the Master Thesis Project has no role in approving or disapproving a student's proposed Specialisation Course. This is simply because students often start their Specialisation Course without already having chosen a thesis supervisor. The thesis supervisor may, however, require passing certain modules as part of the thesis research project.

MSE Specialisation Materials in Engineering Applications

Introduction 1

Materials in Engineering Applications (MEA)

The automotive, offshore, maritime, medical and various other industries all depend on advanced materials technologies. Indeed, innovations in these fields often require materials that must meet contradictory, nearly impossible demands. In this Specialisation Course you will learn how to optimise materials for engineering applications by considering properties with respect to aspects such as production, joining, corrosion and failure. Alternatives for materials processing and the effect of this processing on properties will be addressed.

This Specialisation Course comprises the following obligatory modules:

-) Science of Failure (MS43100)
-) Corrosion Engineering (MS43120)

Furthermore at least three of the following modules should be taken:

-) Determination of Microstructures (MS43801)(You will also need Advanced Characterisation for this course)
-) Joining Technologies (MS43110)
-) Advanced Characterisation (MS43210)
-) Materials Selection for Engineering Applications (MS43115)

Finally, up to 20 EC of elective modules should be chosen.

Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MEA Obligatory fixed (6 ECTS)
--

MS43100	Science of Failure	3
Responsible Instructor	Dr. V.A. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	ME Department Materials Science & Engineering	

MS43120	Corrosion Engineering	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Co-responsible for assignments	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	corrosion, corrosion principles, corrosion types, corrosion prevention and protection	
Course Contents	- Relevance of corrosion - Costs to society - Definitions and electrochemical character of corrosion - General corrosion vs local forms of corrosion - Electrochemical Thermodynamics - Electrochemical Kinetics - Passivity - Galvanic Corrosion and intergranular corrosion - Pitting and Crevice Corrosion - Protection against corrosion	
Study Goals	The student is able to describe the electrochemical nature of corrosion processes, in his professional life to understand the risks hazards and costs due to corrosion phenomena and act upon it in making decisions on metal applications and to make an argumentative selection of materials classes (steel, stainless steel, aluminium alloys, copper alloys) for given applications. More specifically, the student is able to: 1. compose and interpret electrode potentials and polarisation diagrams for a corroding metal or combination of metals 2. calculate the corrosion rate of metals from electrochemical data set 3. identify and explain the mechanisms of the most important corrosion phenomena 4. select and discuss the technical best protection measures to prevent specific corrosion failures 5. criticize and judge a given materials application	
Education Method	Lectures and guest lectures	
Literature and Study Materials	- Principles and Prevention of Corrosion, Second Edition 1996, van Denny A. Jones, ISBN 0133599930 - lecture slides made available after the lectures	
Assessment	The examination will be as follows: group assignment (including report and presentation) and an individual closed book written exam. In case of unforeseen circumstances, the prescribed assessment will be: group assignment and a remote open book with several fraud prevention measures.	
Enrolment / Application	Registration: More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MEA Obligatory selection (Select at least 8 ECTS)

MS43110	Joining Technologies	3
Responsible Instructor	Dr.ir. M.J.M. Hermans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course will address a range of joining technologies for different material classes (ceramics, polymers, metals), examining the major features, limitations and applications of selected material process combinations. Attention will be paid to the influence of joining on material properties and the influence of the joint on the structure (where applicable). For the chosen case studies, the selection of joining method and consequences in terms of production and lifetime performance will be assessed and associated failure behaviour will be described.	
Study Goals	At the end of this course, students will be able to: describe the major joining methods available select appropriate joining technique(s) based on the material and application requirements assess the influence of a joining technique on the materials involved describe the main failure mechanisms associated with the different joining techniques considered gain familiarity with working in a team and developing a structured approach to addressing manufacturing issues from a materials science perspective gain experience with written and oral communication skills	
Education Method	Lectures from subject experts and lectures prepared by course participants.	
Assessment	Presented lectures and oral examination. Students are required to obtain a minimum mark of 6 for both their lecture and the examination to pass the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43115	Materials Selection for Engineering Applications	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Materials in Design - Evolution of Materials Engineering - The Design process - Engineering materials and their properties - Materials selection charts - Materials selection: case studies; Materials processing and design - Materials Processing Selection Charts - Case studies: process selection - Cost considerations - Environmental Impact - Data sources.	
Study Goals	The student is able to develop a systematic procedure for selecting materials and processes with best matches to the requirements of a design More specifically, the student is able to: - Integrate materials selection with other aspects of design - Develop materials indices and/ or value functions to model design requirements - Identify the main materials engineering aspects of materials and process selection. - Apply all of the above in case studies representing simplified and real designs.	
Education Method	Lecture	
Computer Use	Ansys Granta Edupack Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996	
Assessment	Lecture notes made available through Brightspace after class Final assignment: Written (remote) report on case study of own choice. This examination format will be independent of unforeseen circumstances.	
Enrolment / Application	Registration: More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Analysis Design Integrated Optimalisation Process Project Project planning / management Research Methods Sustainability Technology	
Department	ME Department Materials Science & Engineering	

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43006 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, gamma ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between ions and solids is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials. Understand how electron microscopes, electron and neutron diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958 D.K. Aswal, P.S. Sarkar, Y.S. Kashyap, (Eds.) "Neutron Imaging", Springer Nature, Singapore, 2022.	
Assessment	The examination will be number of assignments including graded peer-review.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. In order to participate in the assessment of the course, participation in the initial classes and the handed out assignments is necessary. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	ME Department Materials Science & Engineering	

MS43315	Recycling Engineering Materials	4
Responsible Instructor	Dr. Y. Yang	
Responsible Instructor	S.T. Abrahams	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials; general physics; Inorganic chemistry; general thermodynamics	
Course Contents	Are you passionate about sustainability and eager to make a real impact on the environment? Dive into the dynamic world of recycling and become a part of the solution to help shape a sustainable future. Why This Course? In today's world, ensuring a sustainable supply of materials while minimizing environmental impact is more crucial than ever. Recycling isn't just about waste management; it's a key pillar of the circular economy, closing the loop in the materials cycle and driving innovation in material production and supply. Who Should Enroll? This course is designed for students with backgrounds in engineering and science who are keen to deepen their understanding of recycling within the context of material production and supply chains. Whether you're looking to innovate in the recycling sector or integrate sustainable practices into your career, this course provides the knowledge and skills you need. The course content is divided into the following topics: Module I: Fundamentals and relevance Introduction: materials recycling as an integrated part of a circular economy Fundamentals of materials recycling Economics of materials recycling Liberation and physical separation technologies Fundamentals of Materials Extraction and Refining Module II: Materials Recycling Technologies Recycling of ferrous base metals Recycling of non-ferrous base metals (Al, Cu, Zn) Recycling of precious and scarce metals Recycling of polymers and plastics: fundamentals and technologies Recycling of polymers and plastics: industrial applications and challenges Recycling of hybrid & composite materials Module III: Recycling of EoL Engineering applications Recycling of e-waste Recycling of end-of-life (EoL) vehicles Recycling of EoL solar PV panels Recycling of rechargeable batteries	
Study Goals	By the end of this course, you will be able to explain the key steps in recycling major engineering materials and design effective recycling routes for complex engineering applications. Specifically, you will: 1. Understand the Materials Cycle: - o Explain the entire value chain of the materials cycle. - o Highlight the role of recycling in closing the loop of this cycle. 2. Master Recycling Technologies: - o Describe how major engineering materials are recycled. - o Identify the best available technologies and current industrial practices. 3. Design Recycling Routes: - o Develop recycling routes for complex products. - o Analyze the benefits, challenges, and limitations of your chosen recycling routes. 4. Evaluate Impacts: - o Assess the economic, social, and environmental impacts of recycling practices.	
Education Method	The course will be given with the following methods: (1) Lectures on campus (2) Guest lectures from industry experts on campus or at the plant site (3) Industrial excursion to a recycling company (4) A case study (group task, to be assigned during the course, incl. mini MOOC online course) All educational resources will be available on the online platform (BrightSpace). They consist of: (1) Compilation of various book chapters and journal/web articles in digital pdf form (under course materials). (2) Lecture presentations (to be delivered after each lecture) (3) Videos and short movies (to be shown during the lectures), with source information given in the lecture presentations. (4) Optional reading materials (not to be included for examinations)	
Assessment	A case study (in small groups) will be performed by all course participants. It will account for 50% of the total evaluation with a report and a presentation in a mini-symposium. The other 50% will be counted with the written exam (closed-book). The assignment is mandatory. Assessment criteria for the assignments are detailed in the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on	

<https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses>.

To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams>.

*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations>.

Department

ME Department Materials Science & Engineering

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE Specialisation Metals Science and Technology

Introduction 1

Metals Science and Technology (MST)

The specialization in Metals Science and Technology will put you in contact with the latest technologies and approaches in the design, characterization, engineering, production and performance of metallic alloys. This specialization will allow you to innovatively design and effectively select metallic alloys able to meet challenging environmental and working conditions. You will earn an in-depth understanding of the relationships between metallic microstructures and their properties, ranging from strength and corrosion to magnetic properties.

This Specialisation Course comprises the following obligatory modules:

-) Science of Failure (MS43100)
-) Metals Science II (MS43200)

Furthermore at least three of the following modules should be taken:

-) Computational Materials Science II (MS43205)
-) Advanced Characterisation (MS43210)
-) Steel Science (MS43215)
-) Corrosion Science (MS43220)

Finally, up to 18 EC of elective modules should be chosen.

Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MST Obligatory fixed (7 ECTS)
--

MS43100	Science of Failure	3
Responsible Instructor	Dr. V.A. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	ME Department Materials Science & Engineering	

MS43200	Metals Science II	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	1. Processes of microstructural evolution: diffusional and displacive phase transformations, precipitation, solidification, production routes. The fundamentals of these microstructural processes will be discussed, as well as specific applications for alloys like titanium, aluminium, nickel-based superalloys, shape-memory, super-elastic and invar alloys. 2. The character of various defects in the crystalline structure of metals and mechanisms by which they are introduced into the structure. Defects in the crystal structure can be present according to thermal equilibrium, but density and character can be affected by heat treatments, plastic deformation, high-energy radiation. 3. The basics of various properties of metallic alloys in relation to crystal defects. Properties of metals are primarily based on the crystal structure, but highly dependent on the density and character of crystal defects. Properties that will be discussed include mechanical behaviour, ferromagnetic properties, shape-memory behaviour, temperature resistance.	
Study Goals	From this course the student will acquire knowledge on: 1. the fundamentals of microstructural processes in metals, 2. the introduction, behaviour and effect of defects in crystalline metallic structures, 3. the relation between crystal defects and properties and performance of metals. More specifically, the student will be able to 1. distinguish the effects of diffusion and interfacial processes in diffusional phase transformations and precipitation, 2. identify the main characteristics of solidification of metals, 3. identify the main characteristics of displacive transformations and the differences with diffusional transformations, 4. describe the implications of these mechanisms for various non-ferro alloys, 5. identify the mechanisms by which defects are introduced into metallic crystals, 6. describe processes by which defects in metals can be controlled, 7. describe the influence of crystal defects on magnetic properties of ferromagnetic materials, 8. describe the essentials of shape-memory behaviour, 9. identify the microstructural processes that determine the temperature resistance of metals.	
Education Method	Lectures in interaction with the student	
Assessment	Written exam, open book	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	
Special Information	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MST Obligatory selection (Select at least 7 ECTS)

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr.ir. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schrodinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Algebra Algorithmics	

	- Analysis - Artificial intelligence - Modelling - Numeric Methods - Programming - Programming concepts - Programming Software - Research Methods - Software - Video
Department	ME Department Materials Science & Engineering

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43006 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, gamma ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between ions and solids is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials. Understand how electron microscopes, electron and neutron diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958 D.K. Aswal, P.S. Sarkar, Y.S. Kashyap, (Eds.) "Neutron Imaging", Springer Nature, Singapore, 2022.	
Assessment	The examination will be number of assignments including graded peer-review.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. In order to participate in the assessment of the course, participation in the initial classes and the handed out assignments is necessary. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	ME Department Materials Science & Engineering	

MS43215	Steel Science	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge regarding solid-solid phase transformations in metals, such as thermodynamics, diffusion, interfaces, nucleation and growth.	
Course Contents	This course is based on assignments, covering mainly the following topics: - Fe-C equilibrium and plain carbon steels - Fe-C-X equilibrium and alloyed steels - Bainite and acicular ferrite - Formation and properties of martensite - Martensite tempering - Thermomechanical processing - Embrittlement and fracture - Simulations at the micro- and atomic-scale for steel research	
Study Goals	To identify the relations between processing, microstructure and properties of steels.	
Education Method	The lecturers are divided in two parts: - A talk, most of the times by an expert guest lecturer. - Discussion on the assignment expected for that day. The self-study component is given by the timely submission of assignments.	
Literature and Study Materials	Different chapters of the book H.K.D.H. Bhadeshia and R.W.K. Honeycombe Steels: Microstructure and Properties , Butterworth-Heinemann, 4th Edition (2017) will be considered. This book is available online at the library. Relevant scientific articles will be provided in Brightspace.	
Assessment	Assignments on provided scientific articles (80% grade) and presentation (20%). It is important to provide the assignments timely, otherwise the grade of the assignment has a penalty, as indicated in Brightspace. Participation in the class discussions is strongly advised and will be included in the final assessment of the course (+/- 1 point of the final grade). There is no exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43220	Corrosion Science	3
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Corrosion is a multidisciplinary phenomenon in which chemistry, physics, and materials are essential. This course covers the fundamental aspects of corrosion science. Students will gain basic scientific knowledge to understand the corrosion mechanisms. The course will discuss the correlation between microstructural features of the metal and the initiation of corrosion, the formation of passive layers, and the role of the environment in corrosion. The most important corrosion mechanisms will be covered. The main barrier protection methods against corrosion and their failure mechanism will be discussed. Following is a list of the main topics addressed in the course: Introduction to the electrochemistry in corrosion: thermodynamic and kinetic Passivity of the metals Localised corrosion Protection methods against corrosion: pre-treatments and coatings Advanced techniques in corrosion research This course will pay great attention to the scientific aspects of corrosion. Students will have the opportunity to know current research topics and their challenges.	
Study Goals	This course aims to discover more fundamental aspects of the corrosion mechanism of metals and protection methods. In this course, the student will be able to: - understand the thermodynamic and kinetic aspects of the corrosion process - gain a basic understanding of electrochemical methods commonly used in corrosion research - analyse and interpret experimental results - understand the effect of the material properties and the environment on the kinetics of the corrosion process - explain the differences in the passivity of iron, aluminium, and stainless steel - identify the microstructural features of the material that influence the initiation of corrosion of relevant metal alloys - understand and describe the mechanism of localised corrosion failures such as pitting failure - read, review and discuss scientific publications in the corrosion field	
Education Method	The lectures will take place on campus. They will consist of theoretical explanations, discussions, and exercises. For some topics, the students must read and prepare material in advance before the session. The assignment will consist of a critical literature mini-review of a research topic selected by the students (with the instructor's guidance). In the review, students will discuss certain numbers of scientific papers in detail and summarise the essential findings and their relevance to the field. The expected outcome of the assignment is a report and a short presentation about their findings. Depending on the number of students in the course, the assignment will be carried out individually or in small groups.	
Literature and Study Materials	The primary study materials are lecture slides and selected scientific papers. Additionally, some selected chapters will be indicated as support reading material. The chapters are from the following books: - Introduction to Corrosion Science by E. McCafferty Springer 2010 - "Corrosion Engineering" by B. Popov - Science Direct 2015 - "Principles and Prevention of Corrosion" by Denny Jones (2nd edition - 1995)	
Assessment	The assessment consists of the Exam (50% of the final grade) and the written Assignment (40% of the final grade) and the Presentation (10 % of the final grade). To pass the course, the student has to obtain a minimum grade of 6 for each part (exam, assignment and presentation).	
Permitted Materials during Tests	The examination will be on campus. The method of assessment will be a close book written exam. In the case of on-campus restrictions, the exam will be an online written test. In this case, the exam will be open book. You can make use of the lecture notes.	
Enrolment / Application	The students will be informed at the beginning of the course. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	
Contact	For general questions or comments about the course, you can contact the docent via email: Dr. Yaiza Gonzalez-Garcia (y.gonzalezgarcia@tudelft.nl).	

MS43330	Sustainable Metals Production	4
Responsible Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	General metals science; Inorganic chemistry, fundamental physical chemistry (thermodynamics, kinetics)	
Course Contents	Main subjects: Primary and secondary metal resources, extractive metallurgy, metals recycling, pyrometallurgy, hydrometallurgy, electrowinning and electro-refining of metals, ironmaking and steelmaking, decarbonization and green transition of steel industry, extraction of major non-ferrous metals (biref), strategic and critical metals, circularity of metals, sustainable development for metals production and recycling, efficient use of energy and materials resource, minimization of environmental footprint (solid & liquid wastes, and CO₂ & other global-warming gas emissions). Lecture topics: The following topics, which are essential for the sustainable metals production, will be covered: Part I: Process fundamentals - theory natural and secondary resources of metals; metals cycle and circularity of metals thermodynamics and kinetics for extractive metallurgy major chemical reactions relevant for the extraction of metals from ore, scrap and waste principles of pyrometallurgy, hydrometallurgy and electrometallurgy (electro-winning, electro-refining, molten slat electrolysis) fundamentals of oxide reduction, sulphide smelting, and pyro-refining technologies, with examples of different metals extraction and refining Part II: Production of metals applications and sustainable developments blast furnace ironmaking, alternative and emerging ironmaking technologies with low CO₂ emissions - direct reduction and smelting reduction, hydrogen-based reduction technologies, electrification and electrolysis basic oxygen furnace (BOF) steelmaking; the electric arc furnace (EAF) steelmaking (scrap-based steel production process) secondary steel refining, continuous casting, steel forming brief introduction of major non-ferrous metals production: (1) alumina and aluminium metal production; (2) the process routes to produce high purity copper from sulphide & low grade oxide ores and scrap copper (pyrometallurgical route, and hydrometallurgical route) (3) hydrometallurgical production of zinc: Roasting Leaching Electrowinning (RLE) route overview on production and recycling of rare and critical metals materials and energy balances, energy and environment, sustainability in metals production	
Study Goals	The students are able to understand and describe the process routes and technologies for producing metals from primary ores and secondary resource (scrap and waste) for the following group of metals: iron and steel, major non-ferrous metals (brief), as well as rare and critical metals (overview). The students can explain the most commonly used methods for producing these metals Pyrometallurgy, Hydrometallurgy, and Electrometallurgy (electrowinning and electrorefining) and understand why these methods are industrially applied, what technologies are involved, which developments are ongoing, and what impacts these processes have on resource utilization, energy consumption and environmental footprint. The students should gain good perception on the technological and societal challenges, and on potential solutions to meeting the future growing demands of metals without compromising the environment.	
Education Method	Lectures and guest lectures from industry experts A case study for metallurgical calculations (thermodynamics and energy consumption in ironmaking, kinetic aspects) and process assessments A plant visit/excursion to a production site for ironmaking and steelmaking (Tata Steel, IJmuiden)	
Literature and Study Materials	Selected book chapters from various monographs (listed below) on primary metals production, and metals recycling (copyright eBook and pdf documents). (1) Peter Hayes, Process Principles in Minerals and Materials Production. 4th edition, 2021. Hayes Publishing Co., Australia. (2) Treatise on Process Metallurgy, Volume 3: Industrial Processes. 2nd Edition, Editor-in-Chief: Seshadri Seetharaman. 2024 Elsevier. Contemporary literature on metals extraction, refining, and recycling Handouts of the lectures in the form of PowerPoint presentations	
Assessment	Assignment: 50% in the form of case study in small groups Written exam: 50% (closed book)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MSE Specialisation Materials for Sustainable Development	
Introduction 1	Materials for Sustainable Development (MSD) This Specialisation Course is about materials science in the context of sustainable development. A growing world population and increasing welfare urge us to redesign our materials world/system towards a more environmental friendly system. We need to close material loops, keep materials longer in the value chain through re-use and remanufacturing, process materials in a resource efficient way, and design out critical/toxic materials. Another important aspect of this specialisation is the energy transition for the production of electricity from hydrocarbon-fuels towards renewable sources and energy efficiency. Energy generation from renewable sources and energy storage is critically dependent on the availability of materials with appropriate properties. This Specialisation Course comprises the following obligatory modules: -) Materials and Sustainable Development (MS43300) -) Materials for Clean Energy Technologies (MS43305) Furthermore, at least two of the following modules should be taken: -) Computational Materials Science II (MS43205) -) Materials at High Temperature (MS43310) -) Advanced Characterisation (MS43210) -) Recycling Engineering Materials (MS43315) Finally, up to 20 EC of elective modules should be chosen. Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MSD Obligatory fixed (8 ECTS)
--

MS43305	Materials for Hydrogen and Solar Energy Technology	4
Responsible Instructor	Dr. A.J. Böttger	
Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSE Specialisation: Materials for Sustainable Development	
Expected prior knowledge	Basic materials science knowledge is required. Basics of semi-conductor physics and photo-voltaics is considered as prior knowledge.	
Course Contents	New technologies require new materials. This course starts with a brief overview on the energy consumption, resource constraints and sustainable alternatives. Within that framework the focus of the course is on materials and devices for solar energy and the production and use of hydrogen as an energy carrier. Materials design, manufacturing and materials behaviour under process conditions is addressed.	
Study Goals	After this course the student is able to: 1. explain and evaluate processes and principles used to produce and store hydrogen 2. explain and evaluate processes and principles used to produce solar energy materials 3. analyse and evaluate the effect of the production methods on the energy conversion properties of solar cell material 4. identify, analyse and evaluate the effects of hydrogen gas on materials 5. identify, analyse and evaluate the effects on materials when solar cells are in use. 6. describe the general design of devices used for conversion of solar radiation and heat into electricity 7. critically read, evaluate and discuss recent scientific publications on the course topics.	
Education Method	Class room teaching and self-study (blend learning) or online alternative + self-study	
Literature and Study Materials	Course material: handouts, videos and/or pre-recorded PowerPoint presentations, recent scientific papers and book chapters.	
Assessment	The examination will be as follows: Written or digital open book exam and an oral presentation + discussion; presence during the oral presentation sessions is obligatory. The oral part of the exam (presentation on recent scientific papers) will take place during the last weeks of the course during regular lecture hours. In case of unforeseen circumstances or measures resulting from COVID-19 or other, the prescribed assessment will be: Written open book exam through online assignment and, oral presentation (on recent scientific papers) + discussion in online sessions (Zoom, Teams or other).	
Permitted Materials during Tests	During the written open book exam the use of course material as provided in Brightspace is permitted. The documents will be made available during the exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Department	ME Department Materials Science & Engineering	
Contact	A.J.Bottger@tudelft	

MS43315	Recycling Engineering Materials	4
Responsible Instructor	Dr. Y. Yang	
Responsible Instructor	S.T. Abrahams	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials; general physics; Inorganic chemistry; general thermodynamics	
Course Contents	Are you passionate about sustainability and eager to make a real impact on the environment? Dive into the dynamic world of recycling and become a part of the solution to help shape a sustainable future. Why This Course? In today's world, ensuring a sustainable supply of materials while minimizing environmental impact is more crucial than ever. Recycling isn't just about waste management; it's a key pillar of the circular economy, closing the loop in the materials cycle and driving innovation in material production and supply. Who Should Enroll? This course is designed for students with backgrounds in engineering and science who are keen to deepen their understanding of recycling within the context of material production and supply chains. Whether you're looking to innovate in the recycling sector or integrate sustainable practices into your career, this course provides the knowledge and skills you need. The course content is divided into the following topics: Module I: Fundamentals and relevance Introduction: materials recycling as an integrated part of a circular economy Fundamentals of materials recycling Economics of materials recycling Liberation and physical separation technologies Fundamentals of Materials Extraction and Refining Module II: Materials Recycling Technologies Recycling of ferrous base metals Recycling of non-ferrous base metals (Al, Cu, Zn) Recycling of precious and scarce metals Recycling of polymers and plastics: fundamentals and technologies Recycling of polymers and plastics: industrial applications and challenges Recycling of hybrid & composite materials Module III: Recycling of EoL Engineering applications Recycling of e-waste Recycling of end-of-life (EoL) vehicles Recycling of EoL solar PV panels Recycling of rechargeable batteries	
Study Goals	By the end of this course, you will be able to explain the key steps in recycling major engineering materials and design effective recycling routes for complex engineering applications. Specifically, you will: 1. Understand the Materials Cycle: - o Explain the entire value chain of the materials cycle. - o Highlight the role of recycling in closing the loop of this cycle. 2. Master Recycling Technologies: - o Describe how major engineering materials are recycled. - o Identify the best available technologies and current industrial practices. 3. Design Recycling Routes: - o Develop recycling routes for complex products. - o Analyze the benefits, challenges, and limitations of your chosen recycling routes. 4. Evaluate Impacts: - o Assess the economic, social, and environmental impacts of recycling practices.	
Education Method	The course will be given with the following methods: (1) Lectures on campus (2) Guest lectures from industry experts on campus or at the plant site (3) Industrial excursion to a recycling company (4) A case study (group task, to be assigned during the course, incl. mini MOOC online course) All educational resources will be available on the online platform (BrightSpace). They consist of: (1) Compilation of various book chapters and journal/web articles in digital pdf form (under course materials). (2) Lecture presentations (to be delivered after each lecture) (3) Videos and short movies (to be shown during the lectures), with source information given in the lecture presentations. (4) Optional reading materials (not to be included for examinations)	
Assessment	A case study (in small groups) will be performed by all course participants. It will account for 50% of the total evaluation with a report and a presentation in a mini-symposium. The other 50% will be counted with the written exam (closed-book). The assignment is mandatory. Assessment criteria for the assignments are detailed in the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on	

<https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses>.

To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams>.

*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations>.

Department

ME Department Materials Science & Engineering

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MSD Obligatory selection (Select at least 6 ECTS)

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr.ir. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schrodinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Algebra Algorithmics	

	- Analysis - Artificial intelligence - Modelling - Numeric Methods - Programming - Programming concepts - Programming Software - Research Methods - Software - Video
Department	ME Department Materials Science & Engineering

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

MS43330	Sustainable Metals Production	4
Responsible Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	General metals science; Inorganic chemistry, fundamental physical chemistry (thermodynamics, kinetics)	
Course Contents	Main subjects: Primary and secondary metal resources, extractive metallurgy, metals recycling, pyrometallurgy, hydrometallurgy, electrowinning and electro-refining of metals, ironmaking and steelmaking, decarbonization and green transition of steel industry, extraction of major non-ferrous metals (biref), strategic and critical metals, circularity of metals, sustainable development for metals production and recycling, efficient use of energy and materials resource, minimization of environmental footprint (solid & liquid wastes, and CO₂ & other global-warming gas emissions). Lecture topics: The following topics, which are essential for the sustainable metals production, will be covered: Part I: Process fundamentals - theory natural and secondary resources of metals; metals cycle and circularity of metals thermodynamics and kinetics for extractive metallurgy major chemical reactions relevant for the extraction of metals from ore, scrap and waste principles of pyrometallurgy, hydrometallurgy and electrometallurgy (electro-winning, electro-refining, molten slat electrolysis) fundamentals of oxide reduction, sulphide smelting, and pyro-refining technologies, with examples of different metals extraction and refining Part II: Production of metals applications and sustainable developments blast furnace ironmaking, alternative and emerging ironmaking technologies with low CO₂ emissions - direct reduction and smelting reduction, hydrogen-based reduction technologies, electrification and electrolysis basic oxygen furnace (BOF) steelmaking; the electric arc furnace (EAF) steelmaking (scrap-based steel production process) secondary steel refining, continuous casting, steel forming brief introduction of major non-ferrous metals production: (1) alumina and aluminium metal production; (2) the process routes to produce high purity copper from sulphide & low grade oxide ores and scrap copper (pyrometallurgical route, and hydrometallurgical route) (3) hydrometallurgical production of zinc: Roasting Leaching Electrowinning (RLE) route overview on production and recycling of rare and critical metals materials and energy balances, energy and environment, sustainability in metals production	
Study Goals	The students are able to understand and describe the process routes and technologies for producing metals from primary ores and secondary resource (scrap and waste) for the following group of metals: iron and steel, major non-ferrous metals (brief), as well as rare and critical metals (overview). The students can explain the most commonly used methods for producing these metals Pyrometallurgy, Hydrometallurgy, and Electrometallurgy (electrowinning and electrorefining) and understand why these methods are industrially applied, what technologies are involved, which developments are ongoing, and what impacts these processes have on resource utilization, energy consumption and environmental footprint. The students should gain good perception on the technological and societal challenges, and on potential solutions to meeting the future growing demands of metals without compromising the environment.	
Education Method	Lectures and guest lectures from industry experts A case study for metallurgical calculations (thermodynamics and energy consumption in ironmaking, kinetic aspects) and process assessments A plant visit/excursion to a production site for ironmaking and steelmaking (Tata Steel, IJmuiden)	
Literature and Study Materials	Selected book chapters from various monographs (listed below) on primary metals production, and metals recycling (copyright eBook and pdf documents). (1) Peter Hayes, Process Principles in Minerals and Materials Production. 4th edition, 2021. Hayes Publishing Co., Australia. (2) Treatise on Process Metallurgy, Volume 3: Industrial Processes. 2nd Edition, Editor-in-Chief: Seshadri Seetharaman. 2024 Elsevier. Contemporary literature on metals extraction, refining, and recycling Handouts of the lectures in the form of PowerPoint presentations	
Assessment	Assignment: 50% in the form of case study in small groups Written exam: 50% (closed book)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE Free Specialisation 14 EC (to be filled in consultation with master coordinator)

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE Recommended electives

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Old collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Dr. K. Masania	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises in class with some additional working material	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
Assessment	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	In this course we introduce the design fundamentals that control specific properties in coatings, surface treatments and functional coatings and the techniques used to apply them and test them. The goal is therefore to provide the fundamentals of chemistry and physics that explain structure-property relationships relevant to understand and design surface treatments/coatings with: (i) all-round relevant properties such as adhesion, mechanical response, active (inhibitors) and passive (barrier) corrosion protection; and (ii) increasingly relevant functional properties such as anticicing, self-cleaning, self-healing, drag, structural colours, storage/release of active chemicals with nano and microparticles, and antifouling. Since material design rules are directly related to application methods, different technologies for coatings application will be covered such as electrophoretic deposition, powder coatings, anodizing, sol-gel and plasma electrolytic oxidation. Characterization strategies to obtain information about key properties will also be discussed. The fundamentals and characterization strategies covered in the course are applicable to coatings, adhesives and composites of interest in aerospace applications and other engineering domains.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Dr.ir. J.A. Pascoe	
Responsible Instructor	N. Eleftheroglou	
Instructor	Ir. J. Sinke	
Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). Regulations applicable to maintenance. - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals		
Education Method		
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam, closed book (60% of final grade) and group assignment (40% of final grade).	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

BM41035	Biomaterials	4
Responsible Instructor	Dr.ir. I. Apachitei	
Instructor	Dr.ir. E.L. Fratila-Apachitei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In improving patient care, especially in relation with tissue loss or dysfunction, the use of proper biomaterials plays a key role. The shift from tissue removal to tissue replacement and at present, tissue regeneration is driven by the (i) evolution of biomaterials from bioinert to bioactive and bioresorbable and (ii) the increasingly complex biomedical problems of an aging and more active population. As an example, the existing total joint replacements have a 75 - 85 % survivability at 15 years. However, since they were first introduced in the 1960s, average life expectancy increased by at least ten years impacting on the quality of the bone and implants survivability that deteriorate with age. As a consequence, improved implant survivability by 10 - 20 years and alternative treatment methods that delay or eliminate the use of prostheses are badly needed. The Biomaterials course covers in an interdisciplinary approach (biomaterial engineering and biology) the background information on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Here, biomaterials and medical devices for orthopedics, cardiovascular system and tissue engineering are covered. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course the students will be able to: 1. Identify biomaterials used in: orthopedics, cardiovascular implants, and scaffolds for tissue engineering 2. Define the properties of biomaterials and their property map (range of property values) 3. Describe structure-property relationships for various biomaterials 4. Describe the methodologies for biomaterials characterization and regulation 5. Compare and contrast biomaterials and biological materials 6. Explain the main interactions between a biomaterial and a biological environment 7. Describe the principles for biomaterials selection and design 8. Describe approaches and processes used for functionalization of biomaterials 9. Identify, analyse and describe a biomaterial problem associated with a medical device 10. Explain the clinical performance of biomaterials and medical devices	
Education Method	Lectures, case studies and workshops	
Literature and Study Materials	Lecture slides and recommended literature (reader)	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Department	ME Department Biomechanical Engineering	

CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3
Responsible Instructor	A.L. Smith	
Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	General chemistry, basics of chemical thermodynamics	
Course Contents	This is a 3 ECTS course composed of 28 total lecture hours, 56 self-study and a final examination. This course is designed for Chemical Engineering, Applied Physics, Materials Science & Engineering, and Sustainable Energy Technology students that are interested in developing a working knowledge of nuclear materials chemistry and the nuclear fuel cycle. The course will cover various aspects of solution and solid state chemistry as well as materials science that play a key role at each step of the nuclear fuel cycle, from the metallurgy of uranium to the disposal of spent reactor fuel or high level waste. While the physics and engineering of controlled fission are central to the generation of electricity by nuclear reactors, chemistry in general and especially materials chemistry dominate all other aspects of the nuclear fuel cycle. This course will give students a comprehensive knowledge of the traditional fuel cycle (the uranium once-through cycle) in Light Water Reactors (LWRs). It will also cover some of the proposed nuclear fuel cycles that may well carry nuclear power through the coming decades. As an outcome of the course, the students will be able to compare and contrast existing and innovative fuel cycles, and discuss the pros and cons of each. In addition, the course will cover some of the unique properties of the actinide 5f-elements. The first three of the actinide series - Th, Pa and U - occur naturally, while the elements following uranium in the Periodic Table (transuranium elements) are man-made. The actinides are essential to nuclear power generation, but also find applications in many other areas of industry, medicine and research. The course will address 5f-electronic configurations, oxidation states, redox potentials, inorganic and structural chemistry, as well thermodynamic properties of actinide compounds. The actinides behaviour in the environment and in the geosphere will be discussed, together with analytical tools for their identification. The biological and environmental hazards associated with these elements pose certain risks which must be controlled and minimized. Due to their significant role and because the final destination of transuranic elements originating from the nuclear fuel cycle is still an open issue, the actinides chemistry and physics continue to be one of the major areas of nuclear research. The materials science part of the course will cover the main features of materials microstructures and behaviour in conditions that are relevant for nuclear reactors. The various types of defects at the atomic or micrometre scale will be explained as well as their consequences in terms of energy. This forms the basis of understanding the changes that materials undergo when subjected to elevated temperature and high-energy radiation. Changes in the microstructure and the defect configurations lead to changes in properties and behaviour and can thus jeopardise the material performance, forming a threat to safety. The course is aimed at master level students with general background in chemistry, and/or materials science.	
Study Goals	1. Students can demonstrate how chemistry and materials science influence almost all aspects of the fuel cycle. 2. Students can describe the key steps involved in the traditional nuclear fuel cycle and explain the involved chemical processes. 3. Students can apply the concepts addressed in the course to discuss alternative fuel cycles, not covered in the class, and explain the fundamental differences with the presented traditional fuel cycle. 4. Students can justify, at each step of the fuel cycle, the behaviour of the different actinide elements in view of their chemical and physical properties (electronic structure, redox potentials, solubility, complexation, speciation, thermodynamics, etc) 5. Students can predict and explain the behaviour of fission products in the irradiated fuel, during the reprocessing and in geological environment. 6. Students are able to predict the environmental behaviour of various actinide species. 7. Students can discuss major analytical techniques for actinide studies. 8. Students can perform numerical evaluations to predict the behaviour of the actinide elements and fission products at each step of the fuel cycle. 9. Students are able to discuss the major features of materials microstructures and behaviour when subjected to the conditions in a nuclear reactor. 10. Students write an essay and make a presentation on a supplemental topic related to the course material, but not covered in class, in which they apply the key concepts presented during the lectures and perform critical analysis of the information.	
Education Method	Oral lectures / Supported self-study	
Literature and Study Materials	Main: Course reader Additional reading: N. Tsoulfanidis. The Nuclear Fuel Cycle, American Nuclear Society, 2013; The Nuclear Fuel Cycle: from Ore to Wastes, P.D. Wilson (ed.), Oxford University Press, 1996. The Chemistry of the Actinide and Transactinide Elements (L.R. Morss, N.M. Edelstein, J. Fuger (Ed.)), Springer Netherlands, 2011 S.A. Cotton, Lanthanide and Actinide Chemistry, John Wiley & Sons, 2006 W. Loveland, D.J. Morrissey, G.T. Seaborg. Modern Nuclear Chemistry, Wiley-Interscience, 2006; G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. Nuclear Fuel Cycle Science and Engineering, I. Crossland (ed.), Woodhead Publishing, 2012 M. Halka, B. Nordstrom, Lanthanides and Actinides, Facts on File Inc., 2011 W. Loveland, D.J. Morrissey, G.T. Seaborg, Modern Nuclear Chemistry, Wiley-Interscience, 2006. G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. William D. Callister Jr., David G. Rethwisch, Materials Science and Engineering: An Introduction, ISBN 978-0470419977, Ed. Wiley	
Assessment	The final grade is composed of the weighted average of the final exam (40%), presentation (20%) and essay (40%). Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Elective	Yes	
Tags	Chemistry	
Location	Building 58 (preferred) or Reactor Institute Delft, Radiation Science and Technology department	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10. In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Prof.dr. I.M.F. Gordon	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

MS43040	Machine Learning for Materials Design	4
Responsible Instructor	Dr. S. Kumar	
Instructor	K.R. Rossi	
Contact Hours / Week x/x/x/x	0.0.4.0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The graduate-level introductory course on "Machine Learning for Materials Design" will explore the principles, methodologies, and applications of machine learning (ML) in the domain of materials science and engineering. By bridging the gap between traditional materials design approaches and the cutting-edge field of machine learning, we will learn the tools and knowledge necessary for data-driven discovery and development of materials. The following ML aspects are first covered through lectures and live coding: - Introduction to PyTorch - Automatic differentiation - Gradient-based optimization - Best practices in machine learning - Feed-forward neural networks - Convolutional neural networks - Recurrent neural networks - Autoencoding neural networks - Inductive biases Next, a wide range of materials science and engineering applications of the above ML topics will be explored via hands-on and latest research-oriented group projects. Some representative examples include (may vary during the course): - Predicting mechanical behavior of composites and polymers - Identifying molecules with unique properties - Designing metamaterials with tailored mechanical behavior - Characterizing imperfections and defects in additive manufacturing through data and physics	
Study Goals	After successfully completing this course, you will be able to: 1. Understand the fundamentals of machine learning and its application in material science. 2. Develop basic machine learning algorithms based on neural networks 3. Identify appropriate machine learning algorithms based on the data and application in material science context 4. Develop data-driven strategies based on machine learning for new problems	
Education Method	Lectures, exercises, and project. Lectures will involve live coding. Students are asked to bring their own laptops.	
Literature and Study Materials	- Lecture slides and Jupyter workbooks will be provided on Brightspace - No specific textbook - Relevant reading material for each section will be provided	
Prerequisites	IMPORTANT: Good knowledge of linear algebra and familiarity with Python are required. Following are the representative topics that are recommended pre-requisites. Mathematics topics: - Vectors and matrices - Tensors - Coordinate transformations - Differentiation (including of matrices) - Eigenvalues and eigenvectors of matrices - Basic knowledge of probability Python topics: - Creating a Function - Creating a Class and Objects - Plotting and visualization of data - Using Numpy for matrix manipulations - Familiarity with jupyter notebooks Materials topics: - Material selection - Mechanical behaviour of materials	
Assessment	Summative assessment: The assessment will be done based on a project, which will have two components: a final presentation and a report. Both final presentation and report will carry 50% of the final grades each. Formative assessment: During the course, there will be several office hours to provide feedback on the project progress. There will also be a mid-term presentation to discuss the project progress. Additional details will be provided at the beginning of the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Algorithmics Artificial intelligence Computer Engineering Databases Design Mathematics Mechanics Modelling Numeric Methods	

	Programming Programming concepts Programming Software Project
Department	ME Department Materials Science & Engineering
Contact	Contact by email: Sid.Kumar@tudelft.nl

MS43100	Science of Failure	3
Responsible Instructor	Dr. V.A. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	ME Department Materials Science & Engineering	

MS43110	Joining Technologies	3
Responsible Instructor	Dr.ir. M.J.M. Hermans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course will address a range of joining technologies for different material classes (ceramics, polymers, metals), examining the major features, limitations and applications of selected material process combinations. Attention will be paid to the influence of joining on material properties and the influence of the joint on the structure (where applicable). For the chosen case studies, the selection of joining method and consequences in terms of production and lifetime performance will be assessed and associated failure behaviour will be described.	
Study Goals	At the end of this course, students will be able to: describe the major joining methods available select appropriate joining technique(s) based on the material and application requirements assess the influence of a joining technique on the materials involved describe the main failure mechanisms associated with the different joining techniques considered gain familiarity with working in a team and developing a structured approach to addressing manufacturing issues from a materials science perspective gain experience with written and oral communication skills	
Education Method	Lectures from subject experts and lectures prepared by course participants.	
Assessment	Presented lectures and oral examination. Students are required to obtain a minimum mark of 6 for both their lecture and the examination to pass the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43115	Materials Selection for Engineering Applications	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Materials in Design - Evolution of Materials Engineering - The Design process - Engineering materials and their properties - Materials selection charts - Materials selection: case studies; Materials processing and design - Materials Processing Selection Charts - Case studies: process selection - Cost considerations - Environmental Impact - Data sources.	
Study Goals	The student is able to develop a systematic procedure for selecting materials and processes with best matches to the requirements of a design More specifically, the student is able to: - Integrate materials selection with other aspects of design - Develop materials indices and/ or value functions to model design requirements - Identify the main materials engineering aspects of materials and process selection. - Apply all of the above in case studies representing simplified and real designs.	
Education Method	Lecture	
Computer Use	Ansys Granta Edupack Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996	
Assessment	Lecture notes made available through Brightspace after class Final assignment: Written (remote) report on case study of own choice. This examination format will be independent of unforeseen circumstances.	
Enrolment / Application	Registration: More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Analysis Design Integrated Optimalisation Process Project Project planning / management Research Methods Sustainability Technology	
Department	ME Department Materials Science & Engineering	

MS43120	Corrosion Engineering	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Co-responsible for assignments	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	corrosion, corrosion principles, corrosion types, corrosion prevention and protection	
Course Contents	- Relevance of corrosion - Costs to society - Definitions and electrochemical character of corrosion - General corrosion vs local forms of corrosion - Electrochemical Thermodynamics - Electrochemical Kinetics - Passivity - Galvanic Corrosion and intergranular corrosion - Pitting and Crevice Corrosion - Protection against corrosion	
Study Goals	The student is able to describe the electrochemical nature of corrosion processes, in his professional life to understand the risks hazards and costs due to corrosion phenomena and act upon it in making decisions on metal applications and to make an argumentative selection of materials classes (steel, stainless steel, aluminium alloys, copper alloys) for given applications. More specifically, the student is able to: 1. compose and interpret electrode potentials and polarisation diagrams for a corroding metal or combination of metals 2. calculate the corrosion rate of metals from electrochemical data set 3. identify and explain the mechanisms of the most important corrosion phenomena 4. select and discuss the technical best protection measures to prevent specific corrosion failures 5. criticize and judge a given materials application	
Education Method	Lectures and guest lectures	
Literature and Study Materials	- Principles and Prevention of Corrosion, Second Edition 1996, van Denny A. Jones, ISBN 0133599930 - lecture slides made available after the lectures	
Assessment	The examination will be as follows: group assignment (including report and presentation) and an individual closed book written exam. In case of unforeseen circumstances, the prescribed assessment will be: group assignment and a remote open book with several fraud prevention measures.	
Enrolment / Application	Registration: More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43200	Metals Science II	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	1. Processes of microstructural evolution: diffusional and displacive phase transformations, precipitation, solidification, production routes. The fundamentals of these microstructural processes will be discussed, as well as specific applications for alloys like titanium, aluminium, nickel-based superalloys, shape-memory, super-elastic and invar alloys. 2. The character of various defects in the crystalline structure of metals and mechanisms by which they are introduced into the structure. Defects in the crystal structure can be present according to thermal equilibrium, but density and character can be affected by heat treatments, plastic deformation, high-energy radiation. 3. The basics of various properties of metallic alloys in relation to crystal defects. Properties of metals are primarily based on the crystal structure, but highly dependent on the density and character of crystal defects. Properties that will be discussed include mechanical behaviour, ferromagnetic properties, shape-memory behaviour, temperature resistance.	
Study Goals	From this course the student will acquire knowledge on: 1. the fundamentals of microstructural processes in metals, 2. the introduction, behaviour and effect of defects in crystalline metallic structures, 3. the relation between crystal defects and properties and performance of metals. More specifically, the student will be able to 1. distinguish the effects of diffusion and interfacial processes in diffusional phase transformations and precipitation, 2. identify the main characteristics of solidification of metals, 3. identify the main characteristics of displacive transformations and the differences with diffusional transformations, 4. describe the implications of these mechanisms for various non-ferro alloys, 5. identify the mechanisms by which defects are introduced into metallic crystals, 6. describe processes by which defects in metals can be controlled, 7. describe the influence of crystal defects on magnetic properties of ferromagnetic materials, 8. describe the essentials of shape-memory behaviour, 9. identify the microstructural processes that determine the temperature resistance of metals.	
Education Method	Lectures in interaction with the student	
Assessment	Written exam, open book	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	
Special Information	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr.ir. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schrodinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Algebra Algorithmics	

	- Analysis - Artificial intelligence - Modelling - Numeric Methods - Programming - Programming concepts - Programming Software - Research Methods - Software - Video
Department	ME Department Materials Science & Engineering

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43006 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, gamma ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between ions and solids is used to characterize the chemistry and microstructure of materials. Describe how the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials. Understand how electron microscopes, electron and neutron diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958 D.K. Aswal, P.S. Sarkar, Y.S. Kashyap, (Eds.) "Neutron Imaging", Springer Nature, Singapore, 2022.	
Assessment	The examination will be number of assignments including graded peer-review.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. In order to participate in the assessment of the course, participation in the initial classes and the handed out assignments is necessary. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	ME Department Materials Science & Engineering	

MS43215	Steel Science	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge regarding solid-solid phase transformations in metals, such as thermodynamics, diffusion, interfaces, nucleation and growth.	
Course Contents	This course is based on assignments, covering mainly the following topics: - Fe-C equilibrium and plain carbon steels - Fe-C-X equilibrium and alloyed steels - Bainite and acicular ferrite - Formation and properties of martensite - Martensite tempering - Thermomechanical processing - Embrittlement and fracture - Simulations at the micro- and atomic-scale for steel research	
Study Goals	To identify the relations between processing, microstructure and properties of steels.	
Education Method	The lecturers are divided in two parts: - A talk, most of the times by an expert guest lecturer. - Discussion on the assignment expected for that day. The self-study component is given by the timely submission of assignments.	
Literature and Study Materials	Different chapters of the book H.K.D.H. Bhadeshia and R.W.K. Honeycombe Steels: Microstructure and Properties , Butterworth-Heinemann, 4th Edition (2017) will be considered. This book is available online at the library. Relevant scientific articles will be provided in Brightspace.	
Assessment	Assignments on provided scientific articles (80% grade) and presentation (20%). It is important to provide the assignments timely, otherwise the grade of the assignment has a penalty, as indicated in Brightspace. Participation in the class discussions is strongly advised and will be included in the final assessment of the course (+/- 1 point of the final grade). There is no exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43220	Corrosion Science	3
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Corrosion is a multidisciplinary phenomenon in which chemistry, physics, and materials are essential. This course covers the fundamental aspects of corrosion science. Students will gain basic scientific knowledge to understand the corrosion mechanisms. The course will discuss the correlation between microstructural features of the metal and the initiation of corrosion, the formation of passive layers, and the role of the environment in corrosion. The most important corrosion mechanisms will be covered. The main barrier protection methods against corrosion and their failure mechanism will be discussed. Following is a list of the main topics addressed in the course: Introduction to the electrochemistry in corrosion: thermodynamic and kinetic Passivity of the metals Localised corrosion Protection methods against corrosion: pre-treatments and coatings Advanced techniques in corrosion research This course will pay great attention to the scientific aspects of corrosion. Students will have the opportunity to know current research topics and their challenges.	
Study Goals	This course aims to discover more fundamental aspects of the corrosion mechanism of metals and protection methods. In this course, the student will be able to: - understand the thermodynamic and kinetic aspects of the corrosion process - gain a basic understanding of electrochemical methods commonly used in corrosion research - analyse and interpret experimental results - understand the effect of the material properties and the environment on the kinetics of the corrosion process - explain the differences in the passivity of iron, aluminium, and stainless steel - identify the microstructural features of the material that influence the initiation of corrosion of relevant metal alloys - understand and describe the mechanism of localised corrosion failures such as pitting failure - read, review and discuss scientific publications in the corrosion field	
Education Method	The lectures will take place on campus. They will consist of theoretical explanations, discussions, and exercises. For some topics, the students must read and prepare material in advance before the session. The assignment will consist of a critical literature mini-review of a research topic selected by the students (with the instructor's guidance). In the review, students will discuss certain numbers of scientific papers in detail and summarise the essential findings and their relevance to the field. The expected outcome of the assignment is a report and a short presentation about their findings. Depending on the number of students in the course, the assignment will be carried out individually or in small groups.	
Literature and Study Materials	The primary study materials are lecture slides and selected scientific papers. Additionally, some selected chapters will be indicated as support reading material. The chapters are from the following books: - Introduction to Corrosion Science by E. McCafferty Springer 2010 - "Corrosion Engineering" by B. Popov - Science Direct 2015 - "Principles and Prevention of Corrosion" by Denny Jones (2nd edition - 1995)	
Assessment	The assessment consists of the Exam (50% of the final grade) and the written Assignment (40% of the final grade) and the Presentation (10 % of the final grade). To pass the course, the student has to obtain a minimum grade of 6 for each part (exam, assignment and presentation).	
Permitted Materials during Tests	The examination will be on campus. The method of assessment will be a close book written exam. In the case of on-campus restrictions, the exam will be an online written test. In this case, the exam will be open book. You can make use of the lecture notes.	
Enrolment / Application	The students will be informed at the beginning of the course. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	
Contact	For general questions or comments about the course, you can contact the docent via email: Dr. Yaiza Gonzalez-Garcia (y.gonzalezgarcia@tudelft.nl).	

MS43305	Materials for Hydrogen and Solar Energy Technology	4
Responsible Instructor	Dr. A.J. Böttger	
Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSE Specialisation: Materials for Sustainable Development	
Expected prior knowledge	Basic materials science knowledge is required. Basics of semi-conductor physics and photo-voltaics is considered as prior knowledge.	
Course Contents	New technologies require new materials. This course starts with a brief overview on the energy consumption, resource constraints and sustainable alternatives. Within that framework the focus of the course is on materials and devices for solar energy and the production and use of hydrogen as an energy carrier. Materials design, manufacturing and materials behaviour under process conditions is addressed.	
Study Goals	After this course the student is able to: 1. explain and evaluate processes and principles used to produce and store hydrogen 2. explain and evaluate processes and principles used to produce solar energy materials 3. analyse and evaluate the effect of the production methods on the energy conversion properties of solar cell material 4. identify, analyse and evaluate the effects of hydrogen gas on materials 5. identify, analyse and evaluate the effects on materials when solar cells are in use. 6. describe the general design of devices used for conversion of solar radiation and heat into electricity 7. critically read, evaluate and discuss recent scientific publications on the course topics.	
Education Method	Class room teaching and self-study (blend learning) or online alternative + self-study	
Literature and Study Materials	Course material: handouts, videos and/or pre-recorded PowerPoint presentations, recent scientific papers and book chapters.	
Assessment	The examination will be as follows: Written or digital open book exam and an oral presentation + discussion; presence during the oral presentation sessions is obligatory. The oral part of the exam (presentation on recent scientific papers) will take place during the last weeks of the course during regular lecture hours. In case of unforeseen circumstances or measures resulting from COVID-19 or other, the prescribed assessment will be: Written open book exam through online assignment and, oral presentation (on recent scientific papers) + discussion in online sessions (Zoom, Teams or other).	
Permitted Materials during Tests	During the written open book exam the use of course material as provided in Brightspace is permitted. The documents will be made available during the exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Department	ME Department Materials Science & Engineering	
Contact	A.J.Bottger@tudelft	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

MS43315	Recycling Engineering Materials	4
Responsible Instructor	Dr. Y. Yang	
Responsible Instructor	S.T. Abrahams	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials; general physics; Inorganic chemistry; general thermodynamics	
Course Contents	Are you passionate about sustainability and eager to make a real impact on the environment? Dive into the dynamic world of recycling and become a part of the solution to help shape a sustainable future. Why This Course? In today's world, ensuring a sustainable supply of materials while minimizing environmental impact is more crucial than ever. Recycling isn't just about waste management; it's a key pillar of the circular economy, closing the loop in the materials cycle and driving innovation in material production and supply. Who Should Enroll? This course is designed for students with backgrounds in engineering and science who are keen to deepen their understanding of recycling within the context of material production and supply chains. Whether you're looking to innovate in the recycling sector or integrate sustainable practices into your career, this course provides the knowledge and skills you need. The course content is divided into the following topics: Module I: Fundamentals and relevance Introduction: materials recycling as an integrated part of a circular economy Fundamentals of materials recycling Economics of materials recycling Liberation and physical separation technologies Fundamentals of Materials Extraction and Refining Module II: Materials Recycling Technologies Recycling of ferrous base metals Recycling of non-ferrous base metals (Al, Cu, Zn) Recycling of precious and scarce metals Recycling of polymers and plastics: fundamentals and technologies Recycling of polymers and plastics: industrial applications and challenges Recycling of hybrid & composite materials Module III: Recycling of EoL Engineering applications Recycling of e-waste Recycling of end-of-life (EoL) vehicles Recycling of EoL solar PV panels Recycling of rechargeable batteries	
Study Goals	By the end of this course, you will be able to explain the key steps in recycling major engineering materials and design effective recycling routes for complex engineering applications. Specifically, you will: 1. Understand the Materials Cycle: - o Explain the entire value chain of the materials cycle. - o Highlight the role of recycling in closing the loop of this cycle. 2. Master Recycling Technologies: - o Describe how major engineering materials are recycled. - o Identify the best available technologies and current industrial practices. 3. Design Recycling Routes: - o Develop recycling routes for complex products. - o Analyze the benefits, challenges, and limitations of your chosen recycling routes. 4. Evaluate Impacts: - o Assess the economic, social, and environmental impacts of recycling practices.	
Education Method	The course will be given with the following methods: (1) Lectures on campus (2) Guest lectures from industry experts on campus or at the plant site (3) Industrial excursion to a recycling company (4) A case study (group task, to be assigned during the course, incl. mini MOOC online course) All educational resources will be available on the online platform (BrightSpace). They consist of: (1) Compilation of various book chapters and journal/web articles in digital pdf form (under course materials). (2) Lecture presentations (to be delivered after each lecture) (3) Videos and short movies (to be shown during the lectures), with source information given in the lecture presentations. (4) Optional reading materials (not to be included for examinations)	
Assessment	A case study (in small groups) will be performed by all course participants. It will account for 50% of the total evaluation with a report and a presentation in a mini-symposium. The other 50% will be counted with the written exam (closed-book). The assignment is mandatory. Assessment criteria for the assignments are detailed in the course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on	

	https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

MS43325	Application of Materials in High Tech Engineering	3
Responsible Instructor	Dr. V.A. Popovich	
Instructor	Dr.ir. M.J.M. Hermans	
Instructor	Dr. E.M. Kelder	
Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. E. Mendes	
Instructor	Dr. H.L. Castricum	
Contact Hours / Week x/x/x/x	0.4.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of Materials Science and Engineering. Preferably, the BSc-course 'Materials Science and Engineering' at ME, TU Delft.	
Course Contents	The course aims at training students in performing material and process selection by using objective and quantitative criteria. A substantial part of the course consists of a compilation of actual cases from and presented by the Dutch High Tech industry (ASML, Ceratec, Euro-Techniek, Hittech, Ramlab, Thermo Fisher, VDL ETG, Tata Steel). TU lecturers subsequently provide further insight into the related materials science background and selection method. Cases encompass: additive manufacturing, steel and metal alloys, titanium, metallic glasses, coatings, ceramics, polymers. Themes: Materials in design - Engineering materials and their properties - High tech applications - Materials selection - Process selection - Case studies - Relation microstructure / technical properties / processing / design - Cost Considerations - Environmental Impact - Data sources.	
Study Goals	Understand the importance of and attain insight into material selection in the design and production of High Tech products, systems and components therein. Acquire the tools to carry out material and process selection based on objective and quantitative criteria. Acquire experience-based knowledge of applications and materials used in High Tech industry. Obtain a firmer knowledge base in the associated material science theory. Acquire adequate basic knowledge for one's future role as an academic engineer in industry: being able to make a good first selection of materials, as well as to understand and apply the advice of materials experts and processing experts. Being able to critically assess both industrial experience and advice from materials overviews and databases.	
Education Method	Case presentations; lectures	
Computer Use	Ansys Granta EduPack Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996 W.D. Callister & D.G. Rethwisch Materials Science & Engineering ISBN:978-1-118-31922-2 Reader background materials science for the case studies of the high-tech industry	
Assessment	Digital exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/faculties/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43330	Sustainable Metals Production	4
Responsible Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	General metals science; Inorganic chemistry, fundamental physical chemistry (thermodynamics, kinetics)	
Course Contents	Main subjects: Primary and secondary metal resources, extractive metallurgy, metals recycling, pyrometallurgy, hydrometallurgy, electrowinning and electro-refining of metals, ironmaking and steelmaking, decarbonization and green transition of steel industry, extraction of major non-ferrous metals (biref), strategic and critical metals, circularity of metals, sustainable development for metals production and recycling, efficient use of energy and materials resource, minimization of environmental footprint (solid & liquid wastes, and CO₂ & other global-warming gas emissions). Lecture topics: The following topics, which are essential for the sustainable metals production, will be covered: Part I: Process fundamentals - theory natural and secondary resources of metals; metals cycle and circularity of metals thermodynamics and kinetics for extractive metallurgy major chemical reactions relevant for the extraction of metals from ore, scrap and waste principles of pyrometallurgy, hydrometallurgy and electrometallurgy (electro-winning, electro-refining, molten slat electrolysis) fundamentals of oxide reduction, sulphide smelting, and pyro-refining technologies, with examples of different metals extraction and refining Part II: Production of metals applications and sustainable developments blast furnace ironmaking, alternative and emerging ironmaking technologies with low CO₂ emissions - direct reduction and smelting reduction, hydrogen-based reduction technologies, electrification and electrolysis basic oxygen furnace (BOF) steelmaking; the electric arc furnace (EAF) steelmaking (scrap-based steel production process) secondary steel refining, continuous casting, steel forming brief introduction of major non-ferrous metals production: (1) alumina and aluminium metal production; (2) the process routes to produce high purity copper from sulphide & low grade oxide ores and scrap copper (pyrometallurgical route, and hydrometallurgical route) (3) hydrometallurgical production of zinc: Roasting Leaching Electrowinning (RLE) route overview on production and recycling of rare and critical metals materials and energy balances, energy and environment, sustainability in metals production	
Study Goals	The students are able to understand and describe the process routes and technologies for producing metals from primary ores and secondary resource (scrap and waste) for the following group of metals: iron and steel, major non-ferrous metals (brief), as well as rare and critical metals (overview). The students can explain the most commonly used methods for producing these metals Pyrometallurgy, Hydrometallurgy, and Electrometallurgy (electrowinning and electrorefining) and understand why these methods are industrially applied, what technologies are involved, which developments are ongoing, and what impacts these processes have on resource utilization, energy consumption and environmental footprint. The students should gain good perception on the technological and societal challenges, and on potential solutions to meeting the future growing demands of metals without compromising the environment.	
Education Method	Lectures and guest lectures from industry experts A case study for metallurgical calculations (thermodynamics and energy consumption in ironmaking, kinetic aspects) and process assessments A plant visit/excursion to a production site for ironmaking and steelmaking (Tata Steel, IJmuiden)	
Literature and Study Materials	Selected book chapters from various monographs (listed below) on primary metals production, and metals recycling (copyright eBook and pdf documents). (1) Peter Hayes, Process Principles in Minerals and Materials Production. 4th edition, 2021. Hayes Publishing Co., Australia. (2) Treatise on Process Metallurgy, Volume 3: Industrial Processes. 2nd Edition, Editor-in-Chief: Seshadri Seetharaman. 2024 Elsevier. Contemporary literature on metals extraction, refining, and recycling Handouts of the lectures in the form of PowerPoint presentations	
Assessment	Assignment: 50% in the form of case study in small groups Written exam: 50% (closed book)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43805	Metallic Materials in Maritime Structures	5
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Dr.ir. M.J.M. Hermans	
Instructor	Dr.ir. R.H. Petrov	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is focused on the application of metals in marines structures. Attention is paid to the structure and properties of steels and light alloys as well as their corrosion resistance. The weldability of high-strength steels is addressed and factors influencing weld joint design and design for welding are considered, together with assessment considerations. The fundamental aspects of corrosion mechanism, corrosion monitoring and control will be introduced. The general outline of the course (number of lectures): - Basics of metal structures, phase diagram, phase transformation of metals (2) - Influence of alloying elements on steel structures (1) - Steel production basics (1) - Rolling technology (2) - Al and Al alloys (1) - Welding process & Solidification (2) - Joining dissimilar materials (2) - Additive manufacturing (1) - Corrosion: Thermodynamic and Kinetics (2) - Corrosion types (1) - Corrosion steels and Light-alloys (1) - Corrosion Protection and Control (1) - Cathodic protection (2)	
Course Contents Continuation	The group assignment will focus on a case selected by the students. Based on the specific requirements of the chosen case, the group must (i) recommend the most appropriate material(s) and production procedure, (ii) assess the risk for mechanical failures and propose possible mitigation technique(s) and (iii) evaluate the corrosion susceptibility and recommend method(s) for protection. Each group will deliver a written report. Example of cases: - A helideck on an offshore platform - The boom of a high lift capacity offshore crane - A new oil pipeline to be laid at 3-km water-depth - An lightweight alloy ship hull - Liquefied natural gas storage tank	
Study Goals	- Describe the main factors governing the structure and properties of metals - Identify joint designs typically employed in fabrication - Determine design factors influencing welded joint design and joint performance - Describe corrosion principles and the correlation with material properties - Identify the main corrosion types frequently found in marine structures - Determine appropriate corrosion protection methods based on their underlying behaviour and mechanisms - Select criteria for corrosion control programs	
Education Method	Lectures and recommended reading	
Literature and Study Materials	The metal fabrication part of the course is addressed through readers. The corrosion part of the course is based on selected book chapters that will be available during the course.	
Assessment	Examination (60%) and Assignment (40%). The exam is in closed-book written format. Students are required to have a minimum score of 6 in both the examination and the assignment to pass this course.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43810	Materials in Art and Design	3
Responsible Instructor	Prof.dr. J. Dik	
Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	Culture, art, archaeology, history, degradation, authenticity	
Course Contents	In recent years materials science is increasingly applied to the study of art and archaeological objects. This field is known as archaeometry and it includes three research directions. First of all, authenticity studies on art objects rely on material analysis of the object. Second, artworks suffer from degradation studies that need to be understood at a materials science level. Third, stylistic developments in art history can often be related to the development of new materials or production techniques. The main challenge in this field is to bridge the interface between the humanities and the sciences. Interpreting the findings of materials science in a non-scientific, historical context is the crucial skill in this course. Besides literature study and lecturing, the course includes excursions to art and archaeological collections.	
Study Goals	The students is able to: - give a general overview of archaeometric approaches in material cultural heritage - identify the archaeometric component of archaeological or art historical research questions - describe relevant analytical techniques in archaeometry - interpret analytical data in the archaeological and art historical context and discuss the findings with specialists from the cultural heritage domain	
Education Method	Lecture	
Assessment	Presentation	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	
Remarks	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations. Old course code: MS4071 Laboratory work: depending on the case different analytical techniques may be applied (light microscopy, electron microscopy, x-ray powder diffraction).	
Department	ME Department Materials Science & Engineering	

MS43825	Radiation Damage in Materials	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0.0.4.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science (e.g. WB2330: Materials Science of BSc-ME ME): point defects, dislocations, grain boundaries, phases, precipitates, thermodynamics and kinetics, diffusion, relationship microstructure and mechanical properties.	
Course Contents	This course describes the effects of irradiation of mainly metals & alloys and explains the mechanisms behind those effects, especially for neutron radiation. The first part explains how damage is created, quantified analytically, and simulated. The differences and similarities between damage created by various particles is discussed. This forms the basic understanding for the physical, microstructural and mechanical changes metals and alloys undergo due to radiation. The 2nd and 3rd part of the courses discusses radiation-induced physical, microstructural and mechanical effects qualitatively and quantitatively. This course is designed for Material Science & Engineering, Applied Physics, and Chemical Engineering students, who are interested in material changes resulting from particle irradiation (mainly neutrons) of materials (mainly metals and alloys) in nuclear and fusion reactors. Lectures will be given by (guest-)lecturers from NRG, DIFFER and TU Delft. Visits to NRG and DIFFER are part of this course. Some of the lectures will be given at NRG and DIFFER.	
Study Goals	1. Students learn the difference in defect morphology between ion and neutron irradiation 2. Students can analytically quantify the amount of damage introduced by various particles 3. Students can describe the key effects, which lead to physical, microstructural and mechanical changes of irradiated materials 4. Students can perform numerical calculations to physical, microstructural and mechanical changes in metals	
Education Method	Oral lectures / Supported self-study / Homework / Site visit	
Literature and Study Materials	Fundamentals of Radiation Materials Science - Metals and Alloys", second edition, by Gary S. Was, Springer 2007, ISBN 978-1-4939-3436-2, ISBN 978-1-4939-3438-6 (eBook), DOI 10.1007/978-1-4939-3438-6	
Assessment	Open book, written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

SET3055	Economics and Regulations of Sustainable Energy Systems	4
Responsible Instructor	Dr. S.T.H. Storm	
Instructor	Prof.dr.ir. L.J. de Vries	
Instructor	G.O. Ndubuisi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed: Introduction to energy (electricity) markets: perfect competition; oligopoly Rational (and irrational) market behaviour; ABM ? Investment decision making; LCOE Opportunity costing / discounting; NPV; IRR; discount rate Markets, (perfect) competition and market failure (esp. externalities & lock-in) Private costs, external cost and social cost; empirical evidence (related to energy) LCOE; discounting; private discount rate; IRR and NPV Social cost-benefit analysis; social discount rate	
Study Goals	After this course students are able to: evaluate and analyse the performance of energy (electricity) markets; identify cases of market failures and design remedial action to rectify these market failures; estimate the LCOE and calculate the NPV and IRR of an electricity investment project; contrast private and social costs of (renewable) electricity generation; assess technological lock-in in energy generation; and appraise the role of public policy in energy markets and in energy technology innovation.	
Education Method	This course consists of lectures. Parallel to the course a simulation game will be played by the students.	
Literature and Study Materials	To be announced.	
Assessment	The final grade of the course consists of the following components: - Individual computer exam (80%) - Report on the simulation game (20%) Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10. Final grade calculation: $(0.8 * \text{computer exam}) + (0.2 * \text{report})$ In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Computer exam: resit computer exam - Report on the simulation: resubmission of the report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Elective	Yes	
Co-Instructor	G.O. Ndubuisi	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

Master Year 2

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MSc Projects and Assignments

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO: K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

The minimum grade for the summative assessment is ≥ 5.8 .

Process grade is 20%

The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

MS43702	Materials Elective Project I	2
Responsible Instructor	Dr. S. Kumar	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This elective course (2 ECs) allows students to study a selected aspect (as a project) of Materials Science and Engineering in greater depth than is covered in the general MSE masters curriculum. The project can be literature-based, laboratory-based, or numerically based and should lead to an enhanced understanding of the selected topic. The project will not be permitted as part of a students course list unless approved a priori. The MSE coordinator (MSEcoordinator-ME@tudelft.nl) will assess the relevance and applicability of the proposed project to the MSE programme and the suitability and academic level of the proposed supervisor. The MSE master coordinator must approve proposals for projects (and supervisors) BEFORE being undertaken by the student. The student will gain experience performing an investigative task and applying a critical attitude to the methodology and outcomes of the work. The project will be assessed based on a scientific written report and will be marked by the project supervisor (MSE staff member) using the corresponding grading matrix. The MSE education coordinator will check and, if necessary, moderate the mark before submission in Osiris. The elective is only open to students following the Materials Science and Engineering master. The elective project can only be done within the MSE department's research lines and facilities. This course cannot be on the same topic as Materials Elective Project I (MS43702) or the External Research Project (MS53200).	
Study Goals	At the conclusion of this project, students should be able to: - apply knowledge at the Materials Science and Engineering master level to the definition solution and discussion of a relevant problem within the framework of the topic selected; - apply relevant knowledge at different levels of abstraction; - show critical analysis of the methods, results and interpretation employed during the project; - communicate effectively regarding solutions to problems, conclusions and knowledge gained with professionals in the field.	
Education Method	Individual work	
Assessment	Grade of the written scientific report Specific grading matrix (available in the Brightspace of Materials Science and Engineering)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS43706	Materials Elective Project II	6
Responsible Instructor	Dr. S. Kumar	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This elective course (6 ECs) allows students to study a selected aspect (as a project) of Materials Science and Engineering in greater depth than is covered in the general MSE masters curriculum. The project can be literature-based, laboratory-based or numerically based and should lead to an enhanced understanding of the selected topic. The project will not be permitted as part of a students course list unless approved a priori. The MSE coordinator (MSEcoordinator-ME@tudelft.nl) will assess the relevance and applicability of the proposed project to the MSE programme and the suitability and academic level of the proposed supervisor. The MSE master coordinator must approve proposals for projects (and supervisors) BEFORE being undertaken by the student. The student will gain experience performing an investigative task and applying a critical attitude to the methodology and outcomes of the work. The project will be assessed based on a written report (or scientific paper) and will be marked by the project supervisor (MSE staff member) using the corresponding grading matrix. The MSE education coordinator will check and, if necessary, moderate the mark before submission in Osiris. The elective is only open to students following the Materials Science and Engineering master. The elective project can only be done within the MSE department's research lines and facilities.	
Study Goals	This course cannot be on the same topic as Materials Elective Project I (MS43702) or the External Research Project (MS53200). At the conclusion of this project, students should be able to: - apply knowledge at the Materials Science and Engineering master level to the definition solution and discussion of a relevant problem within the framework of the topic selected; - apply relevant knowledge at different levels of abstraction; - show critical analysis of the methods, results and interpretation employed during the project; - communicate efficiently in the style of a journal or conference article.	
Education Method	Individual work	
Assessment	Grade of the written scientific report or paper in the style of a published article Specific grading matrix (available in the Brightspace of Materials Science and Engineering)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

MS53200	External Research Project	15
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The External Research Project involves a research assignment performed individually during which the student works (more or less) full-time in an external research environment, whether academic or industrial. The assignment is at a level close to that expected during the Masters research project. In this way, the student learns to analyse a given issue, devise an approach toward a solution, set up and perform measurements or calculations and discuss and conclude from the results. A report is written that describes and discusses the work performed. By performing the External Research Project, the student learns to: - define, limit, solve and discuss problems within the framework of a predefined project both independently and in consultation with others; - report on the work performed - gain experience in an organisation other than TU Delft. This project must be on a different topic than MS43702 Materials Elective Project I or MS43706 Materials Elective Project II. The course is only open to students studying within the Materials Science and Engineering Department.	
Study Goals	At the conclusion of this study, the student is able to: - utilise a broad and profound knowledge of Materials Science and Engineering in the definition solution and discussion of a relevant problem within the framework of a project addressing engineering sciences (mathematics, physics and chemistry); - apply relevant knowledge at different levels of abstraction, including a reflective understanding of relationships to other fields. - apply knowledge of paradigms, methods and tools, to critically analyse, model, simulate, and/or design in the performance of the research concerning the defined problem. - demonstrate capabilities to solve technological problems systematically, involving critical problem analysis, formulating sub-problems and providing innovative technical solutions in new and unfamiliar situations. - work independently and/or in multidisciplinary teams (as the situation demands), interacting with specialists and taking the initiative where necessary. - communicate effectively regarding solutions to problems, conclusions, and knowledge gained with professionals in the field.	
Education Method	Individual research project	
Practical Guide	The contact between the student and host institution can be initiated by either the student, the host institution, the External Research Project coordinator or another academic staff member. The host institution agrees on an assignment with the student that requires an academic approach on the level reached after the first year of the Masters programme. A brief assignment description is submitted to the coordinator beforehand for approval. The responsibility for the assignment and the supervision is with one of the lecturers of Delft University of Technology, chosen by the student and the coordinator in relation to the subject of the assignment. The host organisation also assigns a supervisor. The student will work most of the time at the host organisation, but, if it is beneficial for the project, part of the time can be spent at Delft University. The External Research Project is finalised by a written report, which is submitted to the supervisor at the host organisation, the supervising lecturer at the university and the coordinator of the External Research Project within four weeks of completion of the practical work. The report should have a typical length of 20 to 40 pages, be written in English and give an account of the approach, the results obtained, a critical discussion of the results and any conclusions drawn. The report should also contain a concise description of the relevance of the project for the host institution. The assessment of the External Research Project is undertaken by the co-ordinator on the basis of indications by the supervisors at the host institute and the supervising lecturer of the university.	
Assessment	Written Report The student will receive a final grade (6 is the minimum passing grade)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Materials Science & Engineering	

Year	2024/2025
Organization	Mechanical Engineering
Education	Master Materials Science and Engineering

MSE MSc Graduation Project (45 ECTS)

Introduction 1

Literature Research and Master Thesis Project

During this final master programme component the individual student undertakes a literature research (10 EC) and performs independent scientific research leading to a thesis (35 EC).

The Master Thesis Project should only be undertaken, if no more than 10 EC from other courses are still to be obtained. Furthermore, before starting the project the student should make sure that one of the staff members of the department Materials Science and Engineering takes responsibility for it and subsequently acts as a thesis supervisor.

The Master Thesis Project may be related to the chosen Specialisation Course, but can also concern another Materials Science and Engineering subject of interest. The research can be undertaken in one of the Materials Science and Engineering research groups, in another research group of Delft University of Technology, outside the university in an Industrial environment, or in collaboration with another educational institute. The thesis has the same emphasis on original, independent research as a PhD thesis, but is of reduced scope. The Master Thesis Project culminates in a presentation and oral defence of the written thesis.

Some examples of research projects are:

- Welding aluminium alloys without causing residual stresses or distortion
- Development of self-healing coatings for turbine blades
- Recycling neodymium-magnet scrap
- Computer simulation of friction and adhesion
- Hydrogen storage for energy applications
- Authenticity of Van Gogh paintings
- Towards new light-emitting polymers
- Micromechanics, fracture mechanics and chemomechanics of concrete
- Thermomechanical fatigue of SiMo cast iron used in truck engines
- Sustainability of building materials and environments

MS53010-20	MSE Literature Research	10
Responsible Instructor	Y. Gonzalez Garcia	
Assistant	S.M. van der Meer	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The purpose of this course is to make a critical assessment of the literature covering a selected aspect of Materials Science and Engineering. The study is undertaken in preparation for the Master's graduation project (MS53035).	
Study Goals	The learning goals of the literature study are: - systematically collect and interpret extensive and relevant quantities of information from different perspectives and sources (industrial, societal and scientific); - summarize the information in a systematic overview representing the State-of-the-Art literature in the field of your MSc thesis and formulate a problem statement/research goal for your MSc Thesis project, if it applies; - take a systematic approach when performing a literature study - perform a thorough gap analysis - If the topic is related to your MSc thesis, identify the most relevant research questions and opportunities in the field of your thesis project Transferable skills you should demonstrate: - a scientific attitude (i.e. independent, critical and reflective); - the ability to provide a sound line of reasoning and to take a substantiated position; - the ability to draw conclusions that point towards scientific opportunities and to formulate research objectives; - a writing style and quality that equals the required level of the MSc thesis (follows the hourglass method) - a responsible and proactive attitude	
Education Method	Individual work. Self-study	
Prerequisites	The student must have approval from the MSE master coordinator BEFORE starting the literature review.	
Assessment	The deliverables of the literature review are: - a written report - a presentation both, to be completed before starting the graduation research project (MS53035). The assessment consists of - Written report to be graded - A 6 is the minimum grade to pass. - Presentation - Pass/Fail Criteria As part of the master thesis procedure, the student must attend and participate in the Q&A of at least ten (10) MSE colloquium. This should be completed before the graduation date	
Department	ME Department Materials Science & Engineering	
Contact	You can contact the MSE master coordinator, Dr. Yaiza Gonzalez-Garcia by email: MSEcoordinator-ME@tudelft.nl	

MS53035	MSE MSc Thesis	35
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Master Thesis Project is the final program component, during which the student performs a literature survey (MS53010-20) and an independent scientific investigation that leads to a thesis (MS53035). This thesis may relate to the specialisation chosen or another specific area of interest. The research project can be undertaken in one of the Materials Science and Engineering research teams, in another research group of Delft University of Technology, outside the university in an Industrial environment, or in collaboration with another educational institute.	
Course Contents Continuation	Some examples of project topics: - From sand to grade silicon for solar energy application - Design of new materials with unprecedented properties by using machine learning - Design of 2D-materials for catalysis application - Study of 3D printing of materials: from microstructure to properties and performance - Bonding and corrosion failure in electronic devices and solar cells - Study of environmental-friendly corrosion inhibitors for the protection of metals - Influence of heat-treatment and alloying elements on the microstructure development of steels - Critical raw materials in the city: Recycling perspectives - Control of the mechanical properties of additively manufactured and functionally graded metal alloys	
Study Goals	The student is able to meet the requirements set in the exit qualifications of the master's program Materials Science and Engineering. More specifically, the student achieves the following qualifications: 1. A broad and profound knowledge of Materials Science and Engineering and the underlying engineering sciences (mathematics, physics and chemistry). 2. The capability to apply this knowledge at an advanced level at different levels of abstraction, including a reflective understanding of its structure and relations to other fields. 3. The capability for innovative contributions to the discipline in the form of new knowledge on materials or the development of new materials. 4. The capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility with regard to sustainability, economy and social welfare. 5. The capability to solve technological problems in a systematic way, involving problem analysis, formulating sub-problems and providing innovative technical solutions, also in new and unfamiliar situations. 6. Professional attitude towards identifying and acquiring lacking expertise, monitoring and critically evaluating existing knowledge, planning and executing research. 7. The capability to work both independently and in multidisciplinary teams. Effective communication with both professionals and a non-specialised public.	
Education Method	Individual work. Self-study Research project	
Prerequisites	Before starting the Master Thesis Project (MS53010-20 & MS53035), the student should make sure that one of the staff members of the department Materials Science and Engineering takes responsibility for the project and subsequently acts as an MSE thesis supervisor. The student MUST obtain the green light from the master MSE coordinator BEFORE starting the literature review (MS53010-20), to comply with the following pre-requisite: has no more than 10ECs left in courses (at the start of the literature review) The student can start the research project (MS53035) when has completed the Literature Review (MS53010-20) or it is close to being finished (approx. 85%), only upon agreement with the MSE supervisor.	
Assessment	The individual student prepares a master's thesis to report on their project. The project work is then concluded through: - a public colloquium by the candidate (max. 30 min) - a defense in front of the MSc graduation committee (closed oral examination) The discussion (defense), presentation, report and performance will be assessed following the corresponding grading matrix (https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren) As part of the master thesis procedure, it is mandatory for the student to attend and participate in the Q&A of at least ten (10) MSE colloquium. This should be completed before the graduation date	
Department	ME Department Materials Science & Engineering	
Contact	You can contact the MSE master coordinator, Dr. Yaiza Gonzalez-Garcia by email: MSEcoordinator-ME@tudelft.nl	

S.T. Abrahami

Department	Team Shoshan Abrahami
Telephone	+31 15 27 86243
Room	34.H-2-280

Prof.dr.ing. R.C. Alderliesten

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Alderliesten
Telephone	+31 15 27 85492
Room	62.NB 0.45

Dr. M.W.E.M. Alfeld

Unit	Mech, Maritime & Materials Eng
Department	Team Matthias Alfeld
Telephone	+31 15 27 88253

Dr.ir. I. Apachitei

Unit	Mech, Maritime & Materials Eng
Department	Biomaterials & Tissue Biomechanics
Telephone	+31 15 27 82276
Room	34.E-3-220

Dr.ir. O.K. Bergsma

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Bergsma
Telephone	+31 15 27 85135
Room	62.NB 0.49

Dr.ir. C. Bos

Unit	Mech, Maritime & Materials Eng
Department	Team Erik Offerman

Ir. B.J.E. de Bruin

Department	Cognitive Robotics
Department	Cognitive Robotics

Dr. A.J. Böttger

Unit	Mech, Maritime & Materials Eng
Department	Team Amarante Bottger
Telephone	+31 15 27 82243

Dr. H.L. Castricum

Department	Materials Science and Engineering
Telephone	+31 15 27 84398
Department	Team Joris Dik
Telephone	+31 15 27 84398

Dr. P. Dey

Unit	Mech, Maritime & Materials Eng
Department	Team Poulumi Dey

Prof.dr. J. Dik

Unit	Mech, Maritime & Materials Eng
Department	Materials Science and Engineering
Telephone	+31 15 27 89571

Dr. N.D. Dogan

Department	Team Neslihan Dogan
-------------------	---------------------

N. Eleftheroglou

Department	Group Eleftheroglou
Telephone	+31 15 27 82051

I. Fernandez Villegas

Department	Design for Sustainability
Telephone	+31 15 27 89745
Room	62.NB 0.35

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Fernandez Villegas
Telephone	+31 15 27 89745
Room	62.NB 0.35

Dr. G.A. Filonenko

Department	Team Georgy Filonenko
-------------------	-----------------------

Department	ChemE/Inorganic Systems Engineering
-------------------	-------------------------------------

Dr.ir. E.L. Fratila-Apachitei

Unit	Mech, Maritime & Materials Eng
Department	Biomaterials & Tissue Biomechanics
Telephone	+31 15 27 89083
Room	34.E-3-220

A.R. Gammon

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 81387

Dr. S.J. Garcia Espallargas

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Garcia Espallargas
Telephone	+31 15 27 81637
Room	62.NB 1.27

Ir. N. Geerlofs

Unit	Mech, Maritime & Materials Eng
Department	Team Erik Offerman
Telephone	+31 15 27 84920

Y. Gonzalez Garcia

Unit	Mech, Maritime & Materials Eng
Department	Team Yaiza Gonzalez Garcia
Telephone	+31 15 27 88369
Room	34.H-2-240

Prof.dr. I.M.F. Gordon

Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 89288

Dr. R.M. Groves

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Groves
Telephone	+31 15 27 88230

Prof.dr.ir. J. Hellendoorn

Department	Cognitive Robotics
Telephone	+31 15 27 89007
Room	34.F-2-440

Unit	Mech, Maritime & Materials Eng
Department	College van Bestuur

Telephone	+31 15 27 89007
Room	34.F-2-440

Dr.ir. M.J.M. Hermans

Unit	Mech, Maritime & Materials Eng
Department	Team Marcel Hermans
Telephone	+31 15 27 82286
Room	34.H-1-300

Prof.dr.ir. O. Isabella

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 81947
Room	36.LB 03.430

C. Kassapoglou

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group De Breuker
Telephone	+31 15 27 81382
Room	62.NB 2.05

Dr. E.M. Kelder

Unit	Technische Natuurwetenschappen
Department	RST/Storage of Electrochemical Energy
Telephone	+31 15 27 83262

Prof.dr.ir. L.A.I. Kestens

Unit	Mech, Maritime & Materials Eng
Department	Team Maria Santofimia Navarro
Telephone	+31 15 27 84353
Room	34.H-3-330

Dr. S. Kumar

Department	Team Sid Kumar
-------------------	----------------

Dr. G.H.J. Langejans

Department	Team Joris Dik
Telephone	+31 15 27 87739
Room	34.H-4-170

Dr. K. Masania

Department	Group Masania
Telephone	+31 15 27 82774

Dr. L. Mazzarella

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 88885
Room	36.LB 03.500

S.M. van der Meer

Unit	Mech, Maritime & Materials Eng
Department	Team Sid Kumar
Telephone	+31 15 27 83976
Room	34.H-2-210

Dr. E. Mendes

Unit	Technische Natuurwetenschappen
Department	ChemE/Advanced Soft Matter
Telephone	+31 15 27 82623
Room	58.D2.320

Prof.dr.ir. J.M.C. Mol

Unit	Mech, Maritime & Materials Eng
Department	Team Arjan Mol
Telephone	+31 15 27 86778
Room	34.H-2-260

G.O. Ndubuisi

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 88895

Dr.ir. S.E. Offerman

Unit	Mech, Maritime & Materials Eng
Department	Team Erik Offerman
Telephone	+31 15 27 82198

Dr.ir. J.A. Pascoe

Department	Group Pascoe
Telephone	+31 15 27 86604
Room	62.NB 0.49

Dr.ir. R.H. Petrov

Unit	Mech, Maritime & Materials Eng
Department	Team Maria Santofimia Navarro
Telephone	+31 15 27 85194
Room	34.H-3-230

Prof.dr. S.J. Picken

Unit	Technische Natuurwetenschappen
Department	ChemE/Advanced Soft Matter
Telephone	+31 15 27 86946
Room	58.D2.380

Dr. V.A. Popovich

Unit	Mech, Maritime & Materials Eng
Department	Team Vera Popovich
Telephone	+31 15 27 89568
Room	34.H-2-200

K.R. Rossi

Department	Team Kevin Rossi
-------------------	------------------

M. Sand

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87885
Room	31.B4.250

Dr.ir. R. Santbergen

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 84425
Room	36.LB 03.500

M.J. Santofimia Navarro

Unit	Mech, Maritime & Materials Eng
Department	Team Maria Santofimia Navarro
Telephone	+31 15 27 88250
Room	34.H-3-250

Ir. J. Sinke

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Sinke
Telephone	+31 15 27 85137

Room	62.NB 0.54
-------------	------------

Dr.ir. M.H.F. Sluiter

Unit	Mech, Maritime & Materials Eng
Department	Team Marcel Sluiter
Telephone	+31 15 27 84922
Room	34.H-4-230

A.L. Smith

Unit	Technische Natuurwetenschappen
Department	RST/Reactor Physics and Nuclear Materials
Telephone	+31 15 27 82106
Room	50.01.01.120

Dr. S.T.H. Storm

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 83548
Room	31.c2.160

Dr. R.A.C.M.M. van Swaaij

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 87259
Room	36.LB 03.410

P. Taheri

Unit	Mech, Maritime & Materials Eng
Department	Team Peyman Taheri
Telephone	+31 15 27 84526

Prof.dr.ir. L.J. de Vries

Unit	Techniek, Bestuur & Management
Department	Energie en Industrie
Telephone	+31 15 27 81137
Room	31.B3.030

Dr. Y. Yang

Unit	Mech, Maritime & Materials Eng
Department	Team Yongxiang Yang
Telephone	+31 15 27 82542
Room	34.H-2-190

Dr. D. Zarouchas

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Group Zarouchas
Telephone	+31 15 27 89527